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FRESHLINE: A Digital Smart Salon Booking and Grooming Platform

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For
ITIS 499
Senior Project
Academic Year 2025-2026-Semester II

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12/05/2026

Abstract

Freshline is a smart salon appointment scheduling and grooming solution that seeks to improve upon the regular salon experience. To this end, it provides an integrated solution for the end-user and the business owner. One integrated solution making it possible for salon owners to run and manage their business operations alongside their customers' finds salons, book appointments, shop for grooming products, and get recommendations. The application was developed using Flutter and Supabase to ensure better scalability, reliability, and performance. Through user testing, the usability and performance of the app were evaluated, and it was found that the app performed well and users were satisfied.

Acknowledgments

Our heartfelt thanks go out to our project supervisor, Dr. Ali Zolait, who has provided us with invaluable guidance and support from the inception of this project till its end. He has made a great contribution towards the formation of the final output of our project. We also thank the Department of Information Systems of the College of Information Technology of the University of Bahrain, which has been the source of all the information we needed for our project.

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Chapter 1

Introduction

Over recent years, the industry of salons and grooming has been developing steadily due to growing awareness of individuals about taking proper care of themselves, as well as the high need for convenient services. However, despite the rapid development, there is a large number of salons, which are still using such traditional tools as visiting and booking by calling. They cause numerous problems and disadvantages, such as long-term waiting, lack of order and organization, as well as difficulties for customers in gaining access to them. Therefore, the necessity for the development of a new tool emerges.

Due to the growing use of mobile technology and digitalization in everyday life, it is now easier than ever to change the format of providing services through making it more accessible. Therefore, the project suggests Freshline, a digital smart salon booking and grooming platform, which will be used to improve customer experience and optimize services provided by salons and grooming companies. By means of this project, a practical digital grooming solution for customers will be developed.

In order to solve the problem and make salons and grooming services more user-friendly, a new digital system will be created. In this way, Freshline will facilitate digital transformation in the industry.

1.1 Problem Statement

Though the salon and grooming industry has seen an increasing level of development within Bahrain, the fact that many service providers continue using traditional booking methods like walk-ins and phone calls, which have been associated with inefficient scheduling, makes the process difficult for users while causing additional problems for salon owners. Users face challenges in finding information about the available services and appointments while also struggling to book an appointment in a timely manner. Moreover, salon owners are facing challenges in managing their business activities due to the absence of an effective booking tool.

Currently, most of the tools in the market fail to deliver an all-rounded experience to customers as they focus on particular needs of users rather than providing a fully functional solution.

1.2 Project Objectives

The Freshline system aims to build a digital platform to improve the customer experience when booking appointments at salons in Bahrain. Besides, it will also be aimed at streamlining business operations for salon owners. Therefore, the application must feature a number of useful users will be able to browse salons, book appointments and purchase grooming products and recommendation from a single platform. Additionally, salon owners will be assisted through the project and will get the means to manage their services, employees, and bookings.

1.3 Relevance/Significance of the project

The significance of the Freshline project is in its consideration of the growing need for digital transformation in the Bahraini salon and grooming industry. With the increased number of people using mobile applications and other digital services, customers expect easier and faster ways of reaching out to salons. The project will help to provide these benefits through its provision of a modern platform that allows improved accessibility and reduced waiting time. Also, the application can aid local salons in organizing and managing their businesses. Finally, the project emphasizes the need for incorporating latest technologies such as AI to create even more advanced solutions.

1.4 Report Outline

This report contains chapters that outline the development process of the Freshline system. The second chapter contains the literature review of previous research as well as the existing products from within the industry. The third chapter includes aspects related to the business aspects of the project such as the feasibility analysis, market analysis, and financial planning. Chapter four contains the information related to project management which includes process model, risk management, and project time frame. Chapter five concentrates on the requirement definition and analysis, system requirements, personas, and system models. The system design and implementation that includes the user interface and the system components is outlined in chapter six. The seventh chapter discusses the testing and evaluation process. It highlights usability findings. The eighth chapter concludes the report.

Chapter 2

Literature Review

2.1 Introduction

The salon and grooming market are among the most rapidly growing industries in the service sector owing to an increase in the importance of personal grooming and changing customer needs (Grand View Research, 2023). In general, traditional salon services required customers to visit the business premises personally or make phone calls to make an appointment. While this was common practice, manual salon booking often resulted in ineffective planning, long waiting times, insufficient transparency, and poor accessibility (Dabholkar & Bagozzi, 2002).

Digital innovations significantly changed the nature of delivering and managing services. With the use of web-based systems and mobile apps, customers can efficiently search for services, book them, and communicate with the service provider in real-time. Besides, modern platforms allow businesses to manage their activities more effectively and improve service quality (Laudon & Laudon, 2020). Currently, many online solutions extend their capabilities and provide additional functions such as shopping in an e-store, membership programs, and recommendations. This chapter will explore changes that occurred during the transition from traditional to digital solutions in the salon industry, review available global and regional platforms, and identify key technologies used in contemporary applications for grooming businesses.

2.2 Traditional and Digital Salon Services

2.2.1 Traditional Booking Methods

Historically, the provision of services in terms of salon and grooming has relied upon manual methods such as walk-ins and calls from clients. The customer either visits the salon without any booking beforehand or communicates with the service provider by making phone calls or through social media platforms. Such a system is relatively easy to implement, cheap, and requires no

complex technical setup, hence its continued popularity among smaller salons and groomers (Zeithaml, Bitner, and Gremler, 2018).

However, such methods tend to be disorganized when it comes to scheduling of appointments. The customer may experience ambiguity about availability, causing long waiting times and congestion at peak hours (Kotler et al., 2019). Also, there tends to be less formalization of communication, leading to confusion on both sides of the client-provider relationship. Therefore, traditional methods will fail to meet the demands of modern-day customers looking for fast and reliable service (Ostrom et al., 2015).

2.2.2 Limitations of Traditional Systems

Despite all of their wide use, traditional appointment booking systems have a number of drawbacks concerning their operation and service delivery. One major problem is the full dependence on human operations, which results in the high likelihood of various errors that may be made by staff members. Examples of such mistakes include double bookings, wrong time assignment, or the inability to record an appointment at all. This may negatively affect both client satisfaction and business reputation (Zeithaml, Bitner and Gremler, 2018).

Additionally, traditional appointment booking does not allow clients to access services at their convenience. They can only book their appointments within normal business hours and cannot confirm their appointment availability online. As for the business, traditional appointment booking implies the lack of digital storage where the salon can keep information about clients and services provided. Consequently, the ability to collect data and evaluate performance is highly restricted, hindering improvements in the quality of services provided (Laudon and Laudon, 2020).

2.2.3 Shift to Digital Booking Systems

Digitalization technologies have greatly influenced the development of services, and this applies to salons and groomers. One of the main advantages of these technologies is booking, especially via apps (Laudon and Laudon, 2020). This type of system gives an opportunity to choose services, desired hours and make appointments immediately regardless of time and place (Chaffey, 2022).

For salon business owners, digital platforms present advanced features for operation management. They include automated schedules, employee's management, notifications and even payments. All of these contribute greatly to operational efficiency (Ostrom et al., 2015). Apart from this, digitalization helps increase engagement since there are options for personalized advice, customers' service history, as well as feedback through online reviews and ratings (Kotler et al., 2019).

2.2.4 Comparison Between Traditional and Digital Systems

An obvious comparison of the two models reveals the considerable benefits that the latter brings thanks to advances in technology. The former is relatively simple but less functional due to its reliance on manual operations and lack of scalability. On the other hand, the latter offers a more efficient system of working that contributes to the customer's experience (Laudon and Laudon, 2020).

Moreover, the new system allows constant communication and information updates between customers and service providers, which was impossible before. Digital platforms allow incorporating such features as notifications and analytics and even integrating different services into one package (Chaffey, 2022).

Below presents a comparison of the key differences between traditional and digital salon systems.

TABLE 2.2.4- 1. TRADITIONAL & DIGITAL SALON

Aspect	Traditional Systems	Digital Systems
Booking Method	Walk-in, phone calls	Mobile apps, online booking
Availability	Limited to working hours	24/7 access
Efficiency	Low, manual process	High, automated system
Customer Experience	Waiting time, uncertainty	Fast, convenient, transparent
Record Management	Manual or none	Digital records and tracking
Notifications	Not available	Automated reminders and alerts
Business Insights	Limited	Data analytics and reports

2.3 Existing Salon Platforms (Global and GCC)

2.3.1 Global Platforms

Some of the global systems have been developed in the past few years to help in salon bookings and management services. They assist in improving customer convenience while providing means for management to facilitate their operations. Examples of popular platforms that are in use in different parts of the world are Fresha, Booksy, and Treatwell.

Most of these platforms contain important functionalities such as online appointment booking, searching for services provided, reviews from clients, and the ability to make payments easily (Laudon and Laudon, 2020). Additionally, some of the platforms have business-related capabilities such as staff management tools and reporting features that are necessary for their operations. The high adoption of global platforms shows how reliant businesses have become on technology (Grand View Research, 2023).

Although the global systems have numerous sophisticated capabilities, their applicability is limited due to the international nature of their development. For example, they cannot address some special requirements of regional users such as linguistic concerns (Ostrom et al., 2015).

2.3.2 GCC and Regional Platforms

Regarding the development of digital service platforms in the Gulf Cooperation Council (GCC), the introduction of such platforms has continued to grow because of the growth in internet connectivity and smartphones (Statista, 2023). While specialized salon platforms are less common in the GCC market than on other markets, a notable rise in digital solution utilization for appointment booking and salon management has taken place.

Regional platforms and adaptations of foreign software are used by businesses across GCC countries. Moreover, the presence of platforms like Middy, which mainly focuses on providing healthcare services, demonstrates how successful appointment platforms can be in the GCC market (Ostrom et al., 2015). The platforms mentioned illustrate how digital appointment solutions have started gaining popularity in GCC countries.

However, despite the existence of many platforms, some challenges are faced by most of them. Such issues as the lack of localization, integration, and advanced services like AI support and personalization show the possibility for development in the GCC market (Chaffey, 2022).

2.3.3 Comparison of Existing Platforms

The analysis of the currently used global and regional platforms shows that there are advantages and disadvantages in each of these types of solution. On the one hand, global platforms usually have more functionalities and work without problems, whereas regional solutions are adapted and focused on local users. At the same time, neither of these types has the complete integration of its functions.

The current solutions tend to be oriented only on the main functions such as appointments and basic business management. Though sometimes some of these platforms include other additional functionalities, such as payment options and customer feedback, no one of these solutions unites

several services within the platform (product commerce, loyalty programs, smart recommendations).

Modern solutions intend to unite different functionalities in one platform. Moreover, the modern systems try to optimize not only business processes but also customers' experience using smart technologies and automation of services, as well as using multi-role access. This approach reflects the future trends in service-oriented software.

This provides a comparison of global platforms, GCC platforms, and the proposed Freshline system according to their key characteristics.

TABLE 2.3.3: COMPARISON BETWEEN GLOBAL, GCC, AND FRESHLINE PLATFORM

Aspect	Global Platforms	GCC Platforms	Freshline
Availability	International	Region-specific	Bahrain-focused
Booking System	Available	Available	Advanced with options
Product E-commerce	Limited	Rare	Integrated
Multi-Role Support	Partial	Limited	Full (customer, owner, employee, admin)
Localization	Limited	Moderate	Strong (Arabic & English)
AI Features	Limited	Not available	AI stylist assistant
Loyalty System	Limited	Rare	Integrated
User Experience	Good	Moderate	Enhanced & personalized
Integration Level	Medium	Low	High (all-in-one platform)

2.4 Features of Modern Salon Systems

2.4.1 Core Booking and Service Features

The modern-day salon management software is designed with efficiency in booking and service management in mind. With these systems, customers are able to search for the salons that offer their services, check the information regarding different services and book an appointment depending on the current availability of the service provider (Laudon and Laudon, 2020). This includes scheduling an appointment within certain hours, booking appointments with particular employees and getting booking confirmations.

Additionally, some systems use automated reminders and alerts for bookings to increase customer attendance at appointments (Chaffey, 2022).

2.4.2 Business and Management Features

In addition to customer-related functions, current salon management platforms provide a wide range of business management tools. The purpose of these is to help salon owners manage their daily activities in a more efficient manner. Common functions may include personnel management, scheduling appointments, tracking revenue, and managing product inventory (Laudon & Laudon, 2020).

These systems facilitate real-time access to data and report generation, enabling the owners to make better decisions after analyzing performance (Ostrom et al., 2015). Using these management tools eases pressure on the management and improves work of the organization.

2.4.3 Customer Experience and Engagement Features

In the current digital era, modern-day applications are more engaged in enhancing the customer experience and keeping users engaged. According to Kotler et al. (2019), users would feel more confident in choosing a service if it had customer reviews and ratings. Additionally, many gaming platforms use loyalty programs and incentive-based rewards schemes to get players to come back again and again and create strong bonds with a loyal customer.

Customization is critical as well. Customization can increase the relevance of products and services which can lead to better engagement from customers (Chaffey, 2022). When services begin to incorporate range of features like these not only do, they appeal to user's feelings but such platforms grow as they retain users for longer.

2.4.4 Integrated and Smart Features

State-of-the-art salon systems have incorporated advanced technologies to provide customers with smart functionality. This includes AI-based assistants, personal recommendation, automation tools (Russell and Norvig, 2021). Powered by Artificial Intelligence, we help the users select the hairstyles or services as per their requirements. One can also automate activities by sending notifications, reminders and managing the workflow. (Laudon and Laudon, 2020)

Additionally, many modern platforms are beginning to adopt an integrated and all-in-one solution to applications. The marriage of the user service booking with the product e-commerce with the payment system with the user management (Chaffey, 2022) is an obvious implementation. This integration improves usability and deliver seamless experiences for customers and providers (Ostrom et al., 2015).

2.5 Emerging Technologies in Grooming Applications

2.5.1 Artificial Intelligence and Smart Assistance

Artificial Intelligence is becoming a significant part of modern service-based applications, facilitating a better interaction with the users, and personalizing it (Artificial Intelligence: A Modern Approach). In grooming applications, AI can be applied to offer intelligent suggestions considering preferences of the user, his previous activities, or general tendencies. For instance, assistants may recommend the most appropriate haircuts, grooming services, and products (Chaffey,2022).

Besides, chatbots that utilize Artificial Intelligence make the experience of using the application better as the system will be able to understand questions asked by the users and give immediate answers, thus minimizing the involvement of humans into the process (Laudon and Laudon, 2020).

Consequently, communication becomes more efficient and customized (Ostrom et al., 2015). Modern solutions, including innovative ones like Freshline, use Artificial Intelligence as well.

2.5.2 Mobile and Cloud-Based Systems

Because of the ubiquity of smartphones users have started using mobile apps for all kinds of services. The mobile first design allows users to access what they need anytime, anywhere. This has great importance for convenience and accessibility in modern applications (Chaffey, 2022). Mobile platforms are also easier and more practical to use because of the features they have, such as the following.

Cloud computing is also a part of enabling these systems. Organizes data storage, expansion, and remote access, meaning complicated local setups are not needed for companies to run (Cloud Computing: Concepts, Technology & Architecture). With cloud-based platforms, companies can manage their operations efficiently while deploying apps that are reliable and can handle numerous users simultaneously (Laudon and Laudon, 2020). A perfect example of the seamless integration of mobile and cloud technologies can be found with a platform like Freshline, which delivers a solid performance through both web and mobile devices.

2.5.3 Automation and Personalization

Another important aspect of current software applications is automation that facilitates the process of working and minimizes efforts. As regards salon solutions, automation can be used in scheduling appointments, sending messages about events, and changing status regarding booking an appointment (Laudon and Laudon, 2020). Thus, automation will help to facilitate operations and decrease the likelihood of mistakes made.

Personalization contributes to increased satisfaction of users since it is related to customizing services and content according to preferences (Chaffey, 2022). Based on data analysis, users are provided with personalized notifications and suggestions that are relevant for them (Kotler et al., 2019). Both personalization and automation make software more user oriented (Ostrom et al., 2015). Solutions that incorporate these features into their structure, including Freshline, seek to offer more personalized experiences to customers.

2.6 Chapter Summary

In conclusion, this chapter examined how the salon and grooming services evolved for a reservation process that initially took place by manual and later developed to the current digital process. The article discusses the challenges faced during the booking process when traditional techniques were in place. Moreover, it also highlights the advantages which resulted from the digital technique. It also identified the various attributes of existing global and GCC platforms and illuminated their integration and localization limitations. Finally, it has discussed the prominent features of the modern-day salon system pertaining to booking management and other aspects as well as the impact of emerging technologies in the process.

Chapter 3

Business Plan

3.1 BUSINESS OVERVIEW

3.1.1 Business Description

Freshline is a smart digital salon booking and grooming application that links customers with salons in Bahrain through a single application. The platform provides a very convenient service that enables users to find salons, book appointments, buy grooming products, and receive personalized AI recommendations. At the same time, it enables salon owners to control and manage their business operations like employees, service catalog, booking, and finance. Meanwhile, the administrators can comprehend the whole system and control platform activities. Freshline aims to eliminate problems that frequently arise at traditional salons like long waiting times, people sitting around with no proper organization in place, and inaccessible services with new-age automatic and easy-to-use. Multiple users can access the platform as customer, salon owner, employee and admin. The key feature of this app is online booking, home service, e-commerce, loyalty program, notifications, and AI. This integrated approach makes Freshline a 360-degree solution with a seamless combination of customer experience and business performance in the grooming industry.

3.1.2 Mission and Vision

Mission:

Freshline aims to make the experience at the salon easy and hassle-free through an online platform that links customers effectively with salons. The scheme focuses on trouble free reservations, unhindered access to services and intelligent suggestions, while at the same time enabling salon owners to run their businesses optimally, enhance productivity and ultimately ensure customer satisfaction by using technology.

Vision:

Freshline aspires to be the leading digital platform for grooming and salon services throughout Bahrain, with the future objective of expanding into the GCC region. The goal is to use cutting edge technologies of artificial intelligence and data driven analytics to transform the conventional way of operating salons into a fully connected digital environment where scheduling, managing and enjoying salon services is quick, customizable and anytime accessible.

3.2 FEASIBILITY STUDY

3.2.1 Swot Analysis

The viability of the Freshline platform can be analysed through a SWOT analysis of the strengths, weaknesses, opportunities and threats of the system. This assessment helps to understand the internal strengths and weaknesses of the platform, as well as the external conditions that will impact its viability in the market.

TABLE 3.2.1: SOWT

Strengths	Weaknesses
• An all-in-one platform that merges appointment scheduling, product purchases, and salon operations management • Accommodates four key user types: customer, salon owner, employee, and administrator • Features an AI powered assistant to offer recommendations and user support • Offers booking options for both in salon visits and home service appointments • Enables real time booking, order tracking, and alert notifications	• The system relies on salons being registered and approved before becoming active • Users must create accounts in order to access all available features • Certain functionalities, such as AI tools and subscriptions, require active user engagement to provide value • Having multiple user roles adds to the systems complexity and the effort needed to manage it
Opportunities	Threats
• Survey results depict strong user demand for booking apps.	• Users continue to depend on conventional in-person booking methods.

• Users want functionalities such as online reservations, product buying, and home services • Opportunity exists to enhance salon management and shorten wait times • Room for future feature expansion, including loyalty programs and AI-driven recommendations	• There is strong competition from established booking applications and social media platforms that offer booking features. • Potential users may be reluctant to transition to a new platform. • User trust is influenced by the systems reliability and performance.
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3.2.2 Business Model Canvas

BUSINESS GOALS AND OBJECTIVES Type of digital business: Service & Marketplace Platform Business objective: • Provide a centralized platform for salon booking and management • Connect customers, salons, employees, and admin in one system • Improve salon operations using digital tools • Enhance customer experience with fast booking and AI assistance • Generate revenue through subscriptions, bookings, and product sales		COMPETITORS Closest competitor: • Fresha • Booksy • Treatwell • StyleSeat How will you be different or better? Freshline differentiates itself by bringing all salons into one platform where users can easily browse and book services. It combines booking, product ordering, and management features in a single system, supports multiple user roles, and offers both in-salon and home services. Additionally, features such as AI assistance, loyalty points, and real-time tracking enhance the overall user experience.
ACQUISITION Keyword search phrase: • salon booking app Bahrain • barber booking app • grooming services app • salon near me booking Visitors/month: 500 - 2000 (initial stage)	BEHAVIOR Available domain name: freshline-app.com Most important use case: Customer books a salon appointment through the application	CONVERSIONS Primary conversion goal: Users complete a booking or product order Revenue stream: • subscription plans • Commission from bookings • Commission from product orders
Social media platform: • Instagram • TikTok • X • Facebook Content mix and frequency: • Daily posts about services and offers • Weekly promotions and discounts • Customer reviews and success stories • Educational grooming content Visitors/month: 1000 – 3000		
CONVERSION EQUATION Primary conversion goal: Booking or order completion 2000 visitors/month * 10 % conversion rate = 20 conversions/month		

Figure 3.2.2-1 Freshline Business Model Canvas

3.2.3 Idea Validation

The matching of features to behavior and the industry movement towards digital service platforms confirms the validity of the Freshline idea. The application offers booking, ordering product and business management in one application, which is kind of the demand nowadays for all-in-one solutions. The variety of user roles on the platform means it can be used by everyone in a real-life situation.

The achievements of the system show that the idea is within the realm of technical feasibility. The fact that there are already similar platforms existing successfully proves that our concept is feasible. However, our idea is more advanced as we have home service, AI, and centralized management. The evidence shows that the idea is real, achievable and has the ability to deliver value in the context.

3.3 Industry analysis

3.3.1 Industry Description and Trends

The grooming and salon industry falls under personal care which is a fast growing market. The personal care industry has seen steady growth in recent years due to rising awareness of personal look and changing lifestyle. As per the industry reports, the beauty and personal care market has a value of several hundred billion dollars, and this market is likely to continue growing at a stable pace. Urbanization, rising income, and increasing grooming demand from both men and women are some factors driving this expansion.

Last year, the impact of digital transformation on industry. Due to easy availability of smartphone and internet service, an online platform has been created where customer can book appointment, browse service, and communicate with business easily. Platforms like Fresha, Booksy, and Treatwell have facilitated this shift by providing digital tools that enhance accessibility and service management.

In addition, the present trend emphasizes increasing emphasis on convenience, personalization, and all-in-one service. More and more, buyers like platforms that provide more than just booking – they also want to purchase products, get real-time updates and receive a recommended tailored

offer. There's an increasing emphasis on features like home service option and subscription model reflecting changing user expectation. As a result, the grooming industry becomes ever-more digital, flexible, and customer-centric in their offer through advancing technology and growing demand.

3.3.2 Competitor Analysis

Criteria	Fresha	Booksy	Treatwell	Urban Company	Freshline
Company Profile	Founded in 2015	Founded in 2014	Founded in 2008	Founded in 2014	Founded in 2026
Country	UK	USA	UK	UAE / India	Bahrain
Brand Positioning	Salon & spa booking platform	Mobile booking for grooming services	Online beauty booking platform	On-demand home services platform	All-in-one salon platform
Revenue Stream	Commission-based	Commission-based	Commission-based	Commission-based	Subscription + commission
Business Model	Marketplace	Marketplace	Marketplace	Service marketplace	Integrated platform
Strength	Easy to use and widely adopted	Strong mobile experience	Large network of salons	Strong home service focus	Fully integrated system (booking + products + AI)
Weakness	Limited advanced features	Limited system integration	No product sales	Not focused only on salons	New platform (early stage)
Platform	Website & App	Website & App	Website & App	App	App
Booking System	Yes	Yes	Yes	Yes	Advanced
Product Sales	Limited	No	No	No	Yes
Multi-role Support	Partial	Partial	Limited	Limited	Full
Home Service	No	Limited	No	Yes	Yes
AI Features	No	No	No	No	Yes
Business Management	Yes	Yes	Yes	Limited	Advanced
Localization (Bahrain)	No	No	No	Partial	Yes
Marketing Channels					
Instagram	Yes	Yes	Yes	Yes	Yes
TikTok	Yes	Yes	Limited	Yes	Yes
Facebook	Yes	Yes	Yes	Yes	Yes
Twitter (X)	Yes	Yes	Yes	Yes	Yes

TABLE 3.3.2- 1 COMPETITOR FRESHLINE ANALYSIS

The table compares Freshline vis-à-vis its competitor for both international and regional platform. It details variations in attributes, corporate strategy, and operational capability. Unlike many competitors who primarily focus on booking service, Freshline offers a more integrated solution that includes product sale, AI function, and full business management. Freshline also offers in-salon and home service option, plus strong local adaptation for the Bahrain market, providing competitive advantage.

1- Fresha

This is a booking platform for spa and salon services. Allows users to manage business via website. This tool is an appointment scheduler, payment processor, and manager of a business. It's utilized by many individuals due to its simple interface and free business model. Nonetheless, it stays focused on booking and payment. Moreover, because it does not include advanced function like AI integration or full system.

2- Booksy

A mobile app focused platform for user to book grooming and beauty service – Booksy. The interface and appointment handling are easy to navigate. While popular, Booksy is not a fully integrated system as it does offer features such as product sale, does not come with AI support, and is not a complete business management tool.

3- Treatwell

Treatwell is an online booking platform mainly operating in Europe that allows you to book for beauty and salon services. Users can schedule appointments with a doctor of their choice. It can only book your product and cannot use it for any other advanced feature which includes AI, product selling, and business management.

4- Urban Company

Urban Company is a service platform operating in the GCC and other region, providing home service such as grooming and salon care. This highlights convenience and on-demand delivery. It does not provide salon management or ticketing and is not customized for photography salons. Integrated booking system, AI assistant, product sale, etc. Advanced management feature is not available.

3.4 MARKETING PLANNING

Before developing the marketing plan for Freshline, a feasibility study and industry analysis were conducted to understand market need and opportunity. The marketing planning process starts with identifying the target market through segmentation and defining user group and their trait. The selection of appropriate marketing channel, setting up a clear goal, and detailing the necessary activities to reach potential users are also covered. The scheme also incorporates measuring up the marketing strategies and future innovation for scalability to make sure that the marketing effort is aligned with the platform growth and expansion aim.

3.4.1 Market Segmentation

Market segmentation divides the market into groups of consumers that share common needs and preferences. These can be demographic, behavioral, preference, and geography. This process will enable us to find out what various groups of users need, and thus make the system more efficient.

The Freshline platform has different users; namely, customer, salon owner and employee. As a result, this market is segmented according to their role and how they use the platform.

The segmentation also observes age, pattern of usage, type of service chosen and location in Bahrain. The image below shows the market section of Freshline.

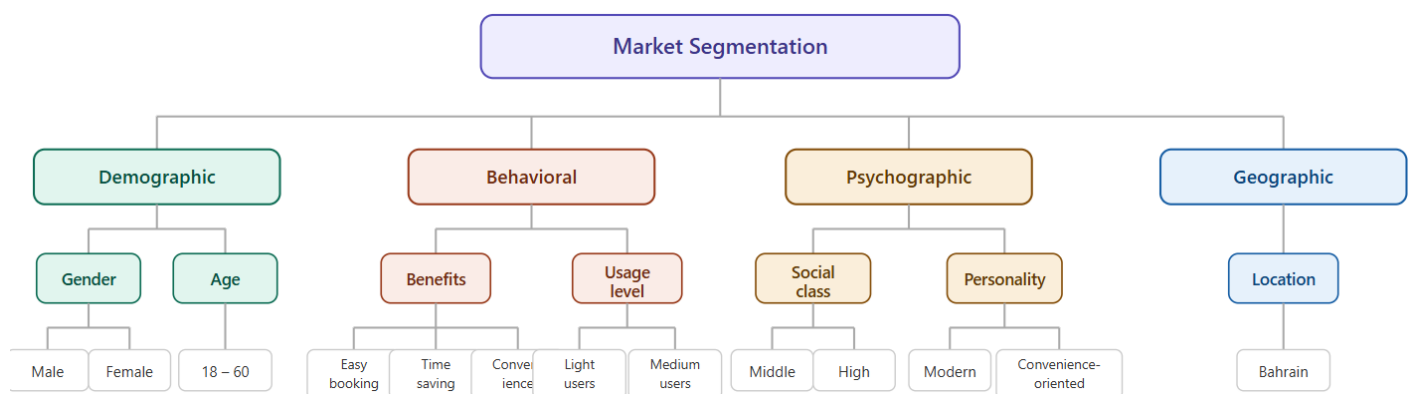


Figure 3.4.1-1 Freshline Market Segmentation

3.4.2 Market Channels

Social media platforms such as Instagram, TikTok, and Twitter (X) rank among the most effective digital tools for audience targeting. Social media services enable users to share content in the form of image, video, and short clip which makes them very powerful for service promotion and customer interaction. For Freshline, these channels will be the primary marketing medium due to its user base in Bahrain and the ability to quickly communicate out to a larger audience. By utilizing the platforms, Freshline can enhance its brand image, promote salon services, and attract new users. Also, users can interact directly with companies through comments, messages, and feedback on social media. These channels are also cheaper than traditional advertising and allow you to do specific targeting based on user interest, location and behaviour. Freshline market share would increase by using social media like Instagram, TikTok, and Twitter.

3.4.3 Marketing Strategy

The goal of marketing strategy is to achieve an important goal set clear marketing target implementation of suitable marketing activity fixing a budget for each one. The initial phase of Freshline will focus on raising awareness and generating user through digital marketing strategy. Through the activity, we will develop a loyal customer for Freshline.

Goal	Objectives	Steps	Budget	Occurrence
1. Increase brand awareness	Influencer marketing	1. Select local influencers 2. Collaborate and promote Freshline	150 BHD	Every 6 months
	Content marketing	1. Create videos and posts 2. Share on Instagram & TikTok	40 BHD	Weekly
	Social media ads	1. Target audience 2. Promote posts	20 BHD	Twice a month
2. Increase app users	Social media growth	1. Create engaging content 2. Post regularly	N/A	Daily
	SEO optimization	1. Use keywords 2. Improve visibility	30 BHD	Monthly
3. Improve customer loyalty	Customer support	1. Respond quickly 2. Solve issues	N/A	Daily
	User feedback	1. Collect feedback 2. Improve system	N/A	Weekly

TABLE 3.4.3-1 FRESHLINE MARKETING STRATEGY

3.5 FINANCIAL PLANNING

The financial plan of Freshline is to estimate the expected revenue and cost in the first year of operation. The objective of this section is to evaluate the financial viability of the platform, thus examining revenue source and operational expense. The sources of income are customer subscription, salon subscription, booking commission, and product sale.

The expense side comprises development and maintenance costs, marketing activity, hosting service, and other operational costs needed to run the platform smoothly. The table presented below gives a month-by-month breakdown of both revenue and expense, which is then grouped into quarterly period.

Category	Jan	Feb	Mar	Q1	Apr	May	Jun	Q2	Jul	Aug	Sep	Q3	Oct	Nov	Dec	Q4	Total
Bronze (4.99 BHD)	150	140	160	450	180	190	200	570	220	230	250	700	300	320	350	970	2,690
Silver (9.99 BHD)	120	110	130	360	150	160	180	490	200	210	230	640	260	280	300	840	2,330
Gold (14.99 BHD)	100	90	110	300	130	140	160	430	180	200	220	600	240	260	280	780	2,110
Salon Monthly (30 BHD)	90	90	120	300	120	150	180	450	180	210	240	630	270	300	330	900	2,280
Salon Yearly (280 BHD)	0	280	0	280	0	280	0	280	0	280	0	280	0	280	0	280	1,120
Booking Commission	50	60	70	180	80	90	100	270	120	130	140	390	160	180	200	540	1,380
Product Sales Commission	40	50	60	150	70	80	90	240	100	110	120	330	140	150	170	460	1,180
TOTAL (BHD)	550	820	650	2,020	730	1,090	910	2,730	1,000	1,360	1,190	3,550	1,110	1,490	1,350	3,950	12,250

TABLE 3.5-1 FRESHLINE REVENUE

The freshline revenue monthly estimate as shown in Table 3.5 1 was done up by quarter. Revenue is generated from customer subscriptions, salon subscriptions, booking commissions, and product sales. As indicated, the revenue rises over time as more users and more salons are expected.

Expense Category	Jan	Feb	Mar	Q1	Apr	May	Jun	Q2	Jul	Aug	Sep	Q3	Oct	Nov	Dec	Q4	Total
App Development & Maintenance	200	150	150	500	150	150	150	450	150	150	150	450	150	150	150	450	1,850
Marketing (Ads & Influencers)	50	50	50	150	70	70	70	210	80	80	80	240	100	100	100	300	900
Hosting & Server Costs	40	40	40	120	40	40	40	120	40	40	40	120	40	40	40	120	480
Operational Costs	60	60	60	180	70	70	70	210	80	80	80	240	90	90	90	270	900
Customer Support	30	30	30	90	40	40	40	120	50	50	50	150	60	60	60	180	540
TOTAL (BHD)	380	330	330	1,040	370	370	370	1,110	400	400	400	1,200	440	440	440	1,320	4,670

TABLE 3.5-2 FRESHLINE EXPENSE

Costs for Freshline include development, marketing, hosting, operation cost, customer support shown in Table 3.5 2. The expense is calculated on a month basis and grouped into quarterly period. The gradual increase over the years resembles that of the platform indicating more services and users need to be supported.

Chapter 4

Project Management

This chapter describes the project management strategy implemented in the design of the Freshline system. It discusses the process model employed in developing the system, the risks which could arise during the project, as well as the project activities schedule. The goal of this chapter is to ensure the successful completion of the project in a timely manner.

4.1 Process Model

Freshline development will follow the Agile SDLC Model. The Agile SDLC Model is an iterative and flexible approach for software development life cycle (SDLC).

Under Agile, development, testing, and enhancements are done continuously in small iterations referred to as sprints or development cycles. In this approach, the development process is divided into several phases, including requirement gathering, designing, implementation, and testing. These phases are done repetitively during development until the system is completely developed. This ensures that the system is evolved continuously, and any changes required are incorporated. Agile approach was chosen for this project due to its flexibility and capability to enhance the system continuously. The system is designed to incorporate several features such as booking, product management, and AI assistance. Therefore, development of these features is done incrementally under the Agile model. This model also enhances collaboration and problem-solving in the development process. This ensures that any issues encountered during the development process are detected and resolved quickly.

4.2 Risk Management

Risk management is essential to identify potential problems that may affect the project. Several risks were identified along with strategies to reduce their impact.

TABLE 4.2-1 RISK MANAGEMENT

Risk	Description	Mitigation Strategy
Technical Issues	Errors in coding or integration	Continuous testing and debugging
Time Constraints	Delay in completing tasks	Follow schedule and prioritize work
Requirement Changes	Changes in user needs	Use Agile flexibility to adapt
System Bugs	Errors during development	Regular testing in each sprint
Data Loss	Loss of important data	Use backup and secure storage

4.3 Project activities Plan

The following section illustrates the project timeline for the creation of the Freshline system. The development of the project began on 1st January 2026 and ended on 12th May 2026. The development phase was broken down into various stages such as requirement gathering, planning, design, implementation, testing, and project closure.

Timeline:

The project timeline provides a general overview of the project phases and their duration across the development period from January to May 2026.

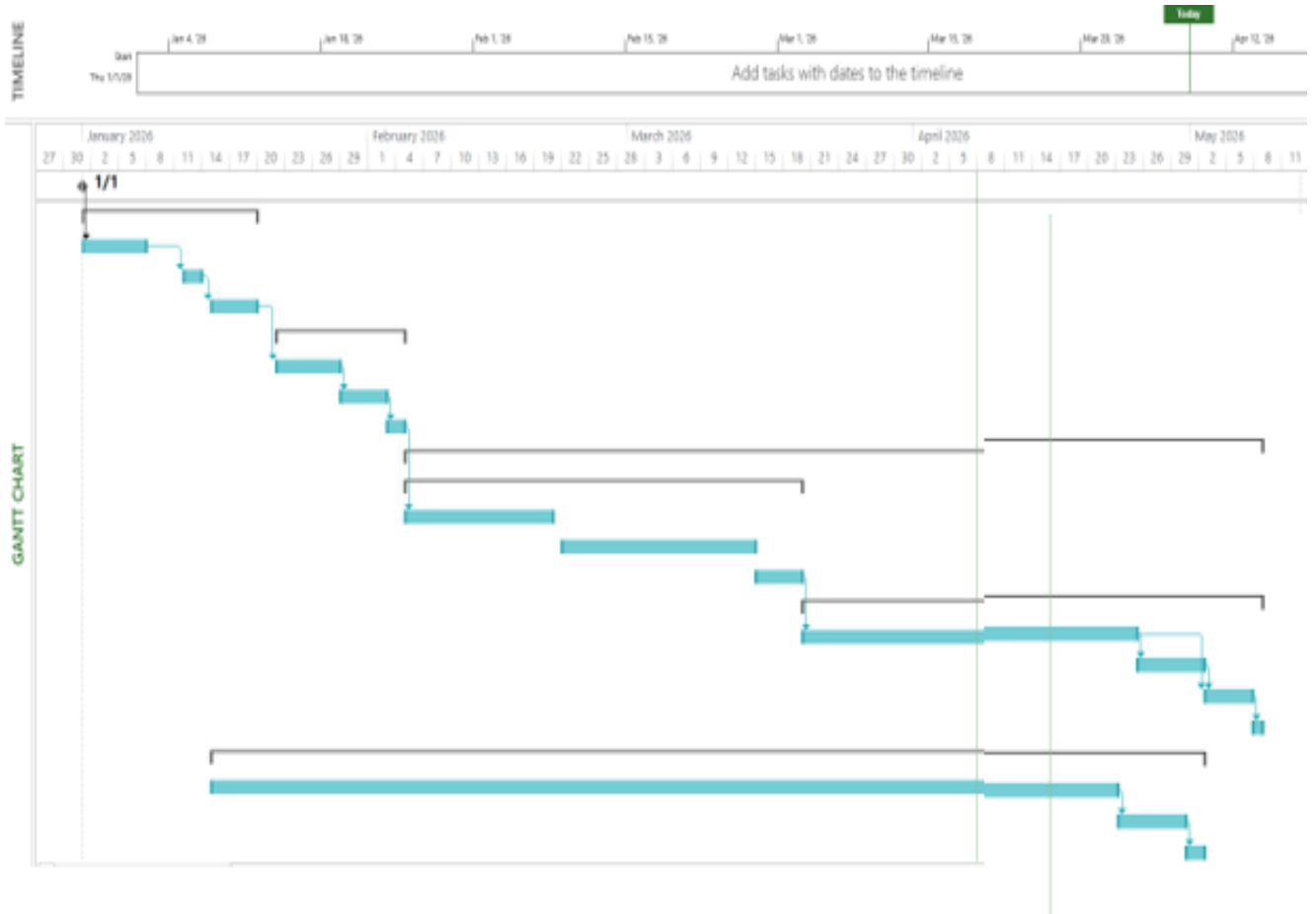


Figure 4.3-1. Timeline

Gantt Chart

The Gantt chart shows all activities of the project, such as the start and end dates, duration, and dependency of each activity. It is helpful in showing the progress of an activity from beginning to end.

ID		Task Mode	Task Name	Duration	Start	Finish	Predecessors
1			Publish project	0 days	Thu 1/1/26	Thu 1/1/26	
2			Project Initiation & Requirements	13 days	Thu 1/1/26	Mon 1/19/26	
3			Define project idea	5 days	Thu 1/1/26	Wed 1/7/26	1
4			Identify User	2 days	Mon 1/12/26	Tue 1/13/26	3
5			Collect requirement	3 days	Thu 1/15/26	Mon 1/19/26	4
6			Planning	10 days	Thu 1/22/26	Wed 2/4/26	
7			Study similar applications	5 days	Thu 1/22/26	Wed 1/28/26	5
8			Analyze system features	3 days	Thu 1/29/26	Mon 2/2/26	7
9			Review technologies	2 days	Tue 2/3/26	Wed 2/4/26	8
10			Execute	60 days	Thu 2/5/26	Wed 4/29/26	
11			Design	31 days	Thu 2/5/26	Thu 3/19/26	
12			System architecture design	12 days	Thu 2/5/26	Fri 2/20/26	9
13			UI/UX design	17 days	Sun 2/22/26	Sat 3/14/26	
14			Database design	5 days	Sun 3/15/26	Thu 3/19/26	
15			Implementation & Testing	29 days	Fri 3/20/26	Wed 4/29/26	
16			Develop frontend (Flutter)	20 days	Fri 3/20/26	Thu 4/16/26	14
17			Backend integration(Supabase)	5 days	Fri 4/17/26	Thu 4/23/26	16
18			Implement main features	3 days	Fri 4/24/26	Tue 4/28/26	16,17
19			Testing and bug fixing	1 day	Wed 4/29/26	Wed 4/29/26	18
20			Monitoring&Controll	71 days	Thu 1/15/26	Thu 4/23/26	
21			Write report chapters	64 days	Thu 1/15/26	Tue 4/14/26	
22			Review and finalize report	5 days	Wed 4/15/26	Tue 4/21/26	21
23			Prepare presentation slides	2 days	Wed 4/22/26	Thu 4/23/26	22
24			Project Closure	0 days	Tue 5/12/26	Tue 5/12/26	

Figure 4.3-2. Gantt Chart

The combination of the timeline and the Gantt chart gives an overview of the project's schedule. As such, while the timeline gives an indication of the general stages in the project, the Gantt chart shows the detailed plan for implementing each activity. In turn, this makes sure that the project is completed successfully.

Chapter 5

Requirement Collection and Analysis

Requirement gathering is an essential process in the development of a software application since it guarantees that the end-product will meet the needs of the users and fit within the overall objectives of the project. This chapter will provide details on requirement identification, analysis, and definition for the Freshline application. Freshline is intended to be a digital platform with a Bahrain-centric approach that will facilitate connections between clients, salon proprietors, staff members, and administrators into one cohesive system. Its purpose is to make it easier for salons to be discovered, appointments made, products purchased, subscription memberships acquired, loyalty points earned, and administration conducted using a single mobile application. Requirement gathering forms the basis upon which the system will be developed.

5.1 Requirement Elicitation

The requirements elicitation for the Freshline software was carried out through undertaking an analysis of the system itself, an observation of market trends and products, and a feature comparison of similar apps for salons and pet groomers. The main techniques used are observing the project objectives and market needs and comparing similar apps for salon booking and e-commerce. Research was done on existing ones to find their similarities, shortcomings and any services missing in terms of appointment availability, house calls, products and services on offer, reward program and administration of the salon itself. We carefully considered the individual requirements of the Bahraini market, which ranged from the phone number format, use of BHD to focusing on Bahrain services. Besides, the target system was evaluated from the perspective of its four major actors' customer, salon owner, employee, administrator ensuring all were catered for specifically. The data collected were critical for specifying system requirements. The proposed system targets the Bahraini Market with four roles of the customers which are Customer, Owner of Salon, Employee and Admin.

5.2 Survey Results

Below is a summary of the key questions and findings from the 81 responses:

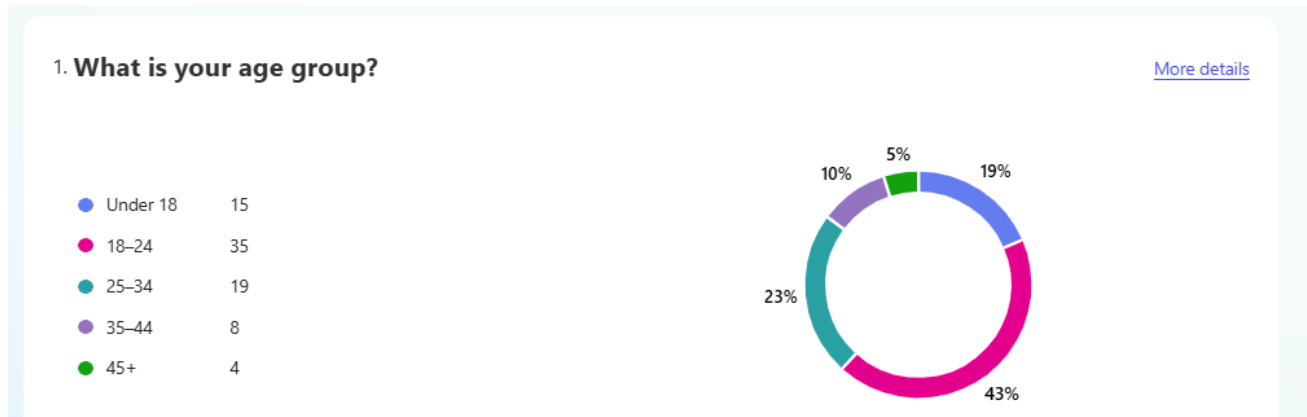


Figure 5.2-1. Age Distribution of Respondents

As can be seen in Figure 5.2-2, respondents reaffirmed their preference for the digital shopping medium owing to convenience and added discounts.

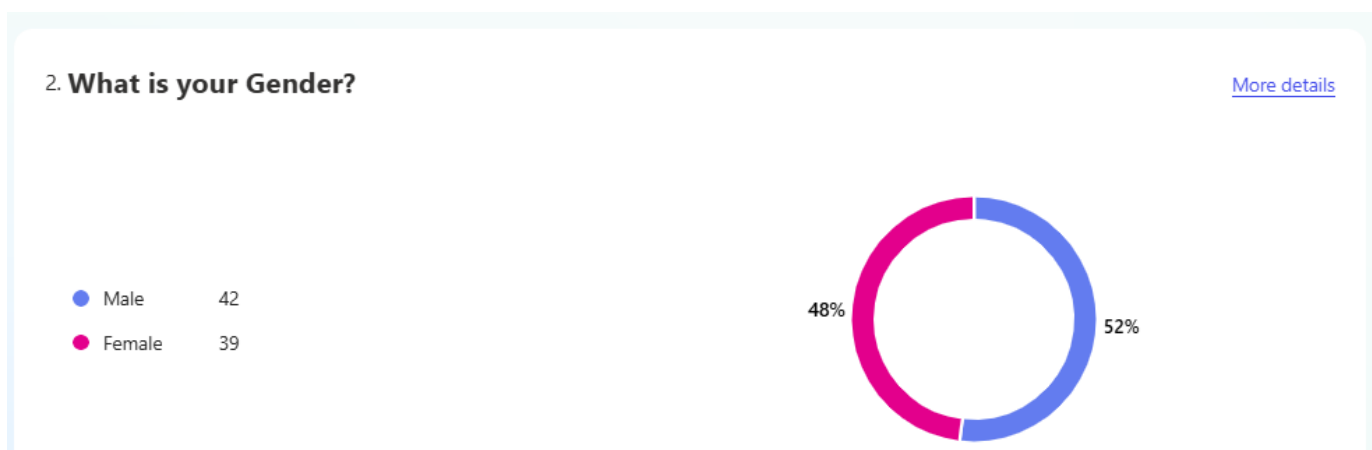


Figure 5.2-2. Gender Distribution of Respondents

As illustrated in Figure 5.2-2, the percentages of male and female respondents are almost equal, where male respondents only slightly ahead at 52% while female respondents make up 48%. In other words, the survey managed to capture the opinion of both genders almost equally.

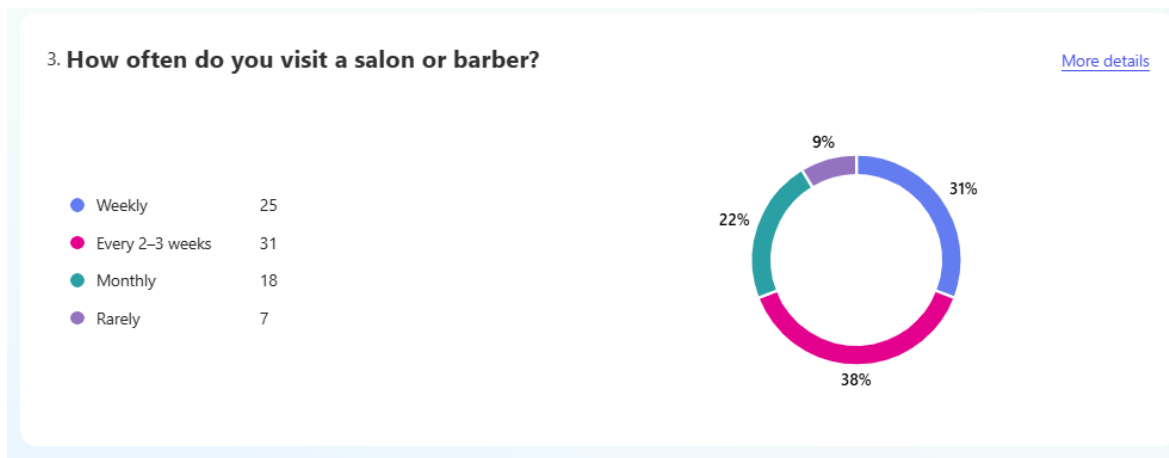


Figure 5.2-3. Frequency of Visiting a Salon or Barber

As illustrated in Figure 5.2-3, the majority of respondents visit the salon or barber shop every 2-3 weeks. This finding indicates that a substantial proportion of users regularly engage in grooming activities and avail themselves of the services offered by salons or barber shops.

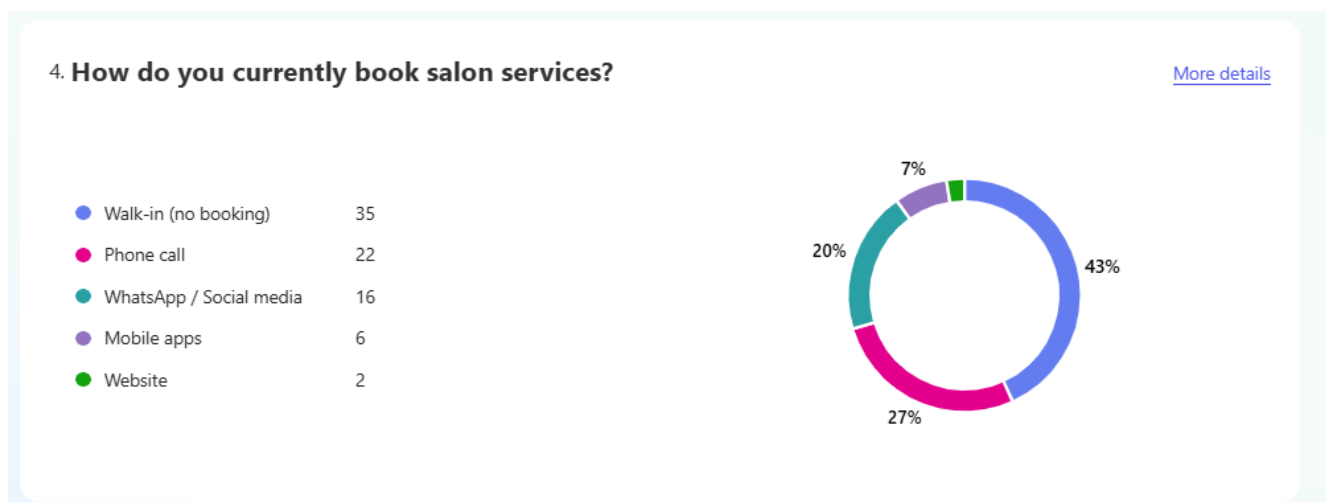


Figure 5.2-4. Methods of Booking Salon Services

As indicated in Figure 5.2-4, a large part of the respondents prefers visiting the service in a walk-in manner (without prior booking). This suggests that traditional means are still in use among users.

5.3 System Requirements



Figure 5.3-1. Problems Faced When Booking Salon Services

As shown in Figure 5.3-1, most respondents had problem in the booking of salon services which indicates a clear demand for an efficient and organized system for booking of salon services.

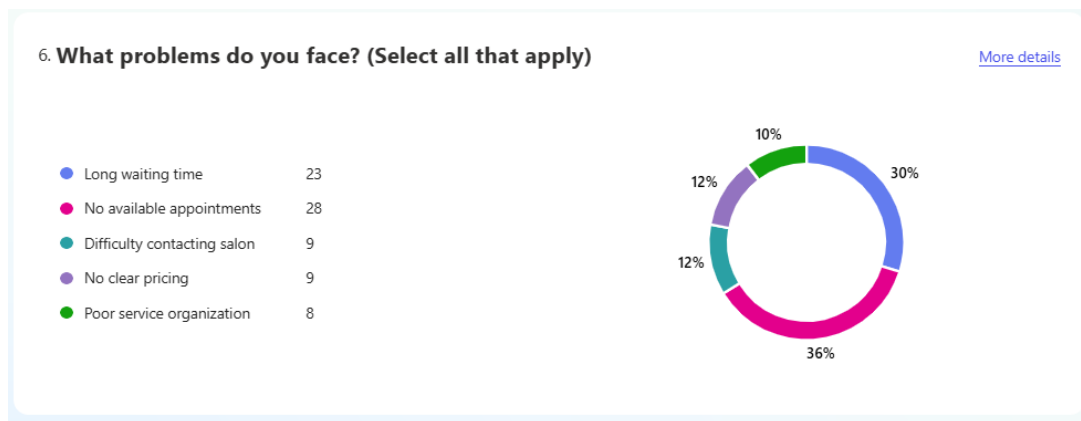


Figure 5.3-2. Problems Faced When Booking Salon Services

According to the data presented in Figure 5.3-2, respondents stated that they faced the issue of availability of credit whereby they cannot get a prior appointment. Following that, lengthy waiting times were mentioned. Thus, it can be seen that there are two fundamental issues that ail the current salon booking systems and appointment confirmation procedure.

7. **Would you use an app to book salon services?**

[More details](#)



Figure 5.3-3. Willingness to Use a Salon Booking Application

As demonstrated in Figure 5.3-3, majority of respondents are interested to opt for salon services using a mobile application. Thus, showing most of the respondents are ready to go for an online option.

8. **What features would you like in such an app? (Select all that apply)**

[More details](#)

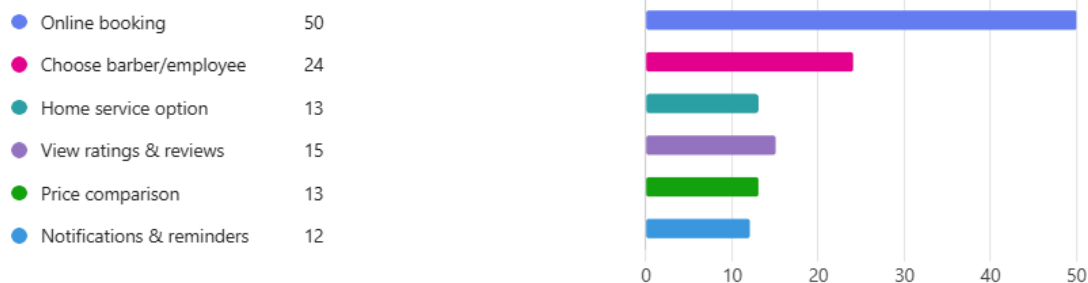


Figure 5.3-4. Desired Features in a Salon Booking Application

Most requested feature is online booking as shown in figure 5.3-4. They also want to choose their barber employee. This mainly helps in making more contacts and bookings with a salon application.

9. Do you buy grooming products (example, hair products, beard care)?

[More details](#)



Figure 5.3-5. Purchasing Grooming Products

As per Figure 5.3-5, majority of respondents buy grooming products which indicates strong interest in integrating this activity in the application.

10. Where do you usually buy them?

[More details](#)



Figure 5.3-6. Preferred Locations for Purchasing Grooming Products

According to the diagram shown as Figure 5.3-6, the majority of respondents prefer buying grooming products from salons indicating the trust in the products provided by the salon rather than the products purchased through other options.

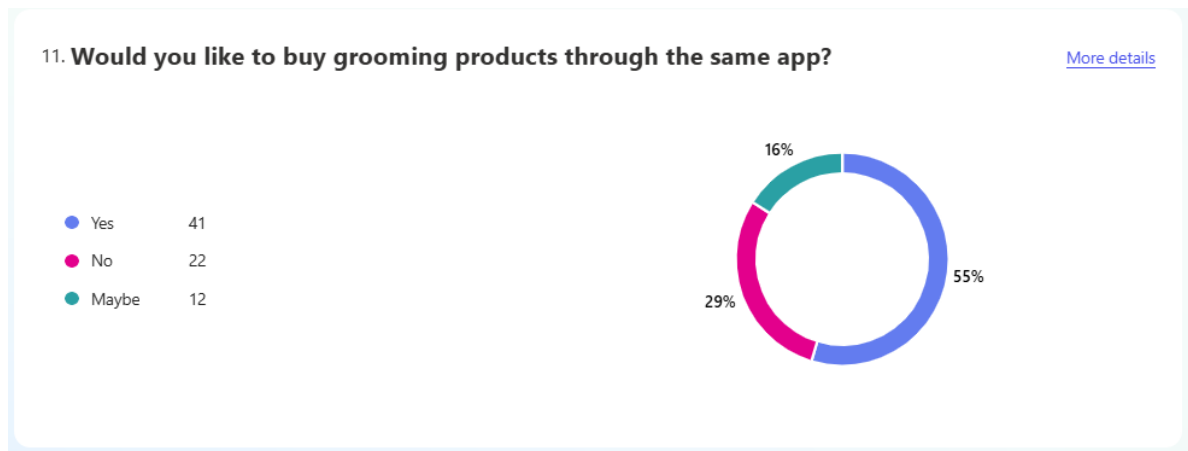


Figure 5.3-7. Willingness to Purchase Grooming Products Through the Application

In figure 5.3-7, it can be observed that most of the respondents expressed their willingness to buy grooming products through the same application. Thus, all these things point towards a very substantial chance of e-commerce in the application.

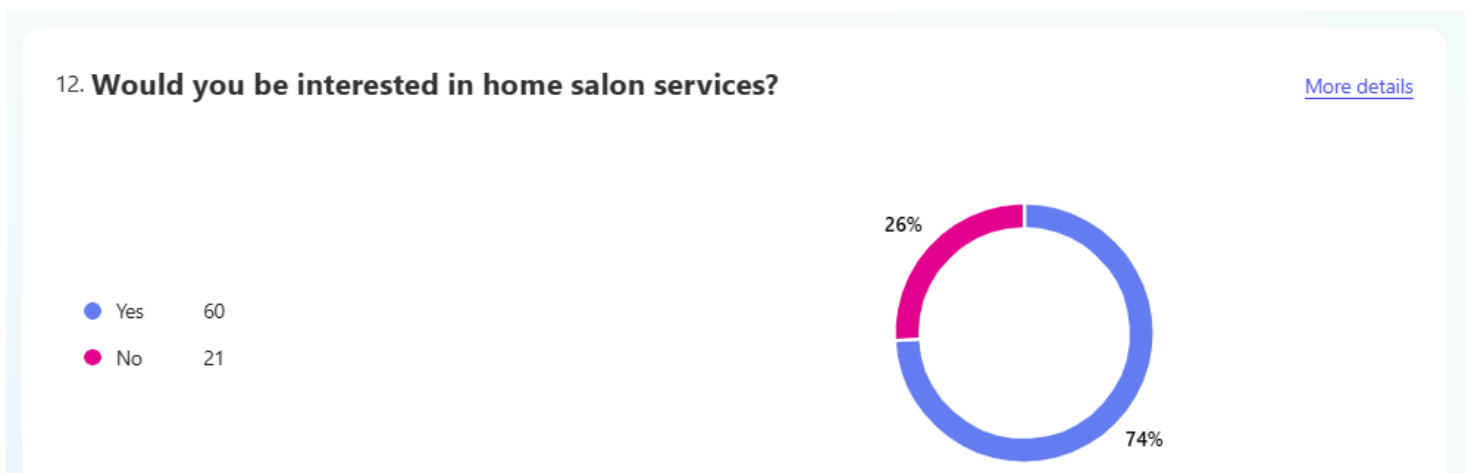


Figure 5.3-8. Interest in Home Salon Services

According to Figure 5.3-8, respondents showed extreme interest in using these services at home, thereby showcasing the need for these services.

13. **Would you use an AI assistant for hairstyle suggestions?**

[More details](#)

● Yes	47
● No	19
● Not Sure	15

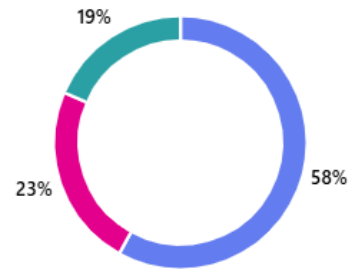


Figure 5.3-9. Willingness to Use an AI Assistant for Hairstyle Suggestions

Responses collected from the users apparently indicate the popularity of the feature of hairstyle suggestions offered by the application as seen in Figure 5.3-9. Most users express willingness to use an AI assistant for this purpose. This indicates that they are interested in having smart and personalized features offered by the application.

14. **Would loyalty points or rewards encourage you to use the app?**

[More details](#)

● Yes	62
● No	19

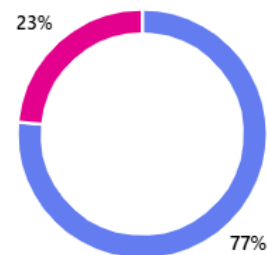


Figure 5.3-10. Impact of Loyalty Points and Rewards on App Usage

As seen in Figure 5.2-14, most respondents stated that they would utilize the application if there were loyalty points and rewards.

15. How useful do you think this app would be?

[More details](#)



Figure 5.3-11. Perceived Usefulness of the Application

Most respondents deem application as very useful based on results as displayed in figure 5.3-11. Indicates a strong positive expectation of application.

16. Any additional suggestions?

29 Responses

ID ↑	Name	Responses	Language
1	anonymous	I think it would be great if you add more services like massages and foot care, and also make package deals instead of just offering things like haircuts and beard trims separately. People usually go for packages more, especially if there are discounts.	English (United States)

Figure 5.3-12. Additional Suggestions from Respondents

5.4 System Requirements

TABLE 5.4-1: FRQ01 – USER REGISTRATION

Field	Description
Requirement ID	FR-01
Requirement Name	User Registration and Login
Requirement Description	The system shall allow users (customers, salon owners, employees, and admins) to register, log in, and manage their accounts securely.
Priority	High
Reason	Authentication is required to access all system features.
Measurement	Users can successfully register and log in using valid credentials.
Requirement Status	Implemented

TABLE 5.4-2: FRQ02 – USER LOGIN

Field	Description
Requirement ID	FR-02
Requirement Name	Salon Discovery
Requirement Description	The system shall allow customers to browse, search, and filter salons based on location, rating, and services.
Priority	High
Reason	Helps users easily find suitable salons.
Measurement	Users can view and filter salons successfully.
Requirement Status	Implemented

TABLE 5.4-3: FRQ03 - SALON BROWSING

Field	Description
Requirement ID	FR-03
Requirement Name	Appointment Booking
Requirement Description	The system shall allow customers to book salon services, select employees, and choose visit mode (in-shop or home service).
Priority	High
Reason	Core functionality of the application.
Measurement	Booking is successfully created and stored in the system.
Requirement Status	Implemented

TABLE 5.4-4: FRQ04 – APPOINTMENT BOOKING

Field	Description
Requirement ID	FR-04
Requirement Name	Product Purchasing
Requirement Description	The system shall allow users to browse, add products to cart, and complete purchases.
Priority	High
Reason	Supports e-commerce functionality within the app.
Measurement	Users can complete product orders successfully.
Requirement Status	Implemented

TABLE 5.4-5: FRQ05 – PRODUCT PURCHASE

Field	Description
Requirement ID	FR-05
Requirement Name	Loyalty Points System
Requirement Description	The system shall provide a loyalty system where users earn and redeem points for discounts.
Priority	Medium
Reason	Encourages user engagement and retention.
Measurement	Points are correctly added and redeemed.
Requirement Status	Implemented

TABLE 5.4-6: FRQ06 – LOYALTY POINTS SYSTEM

Field	Description
Requirement ID	FR-06
Requirement Name	Notifications
Requirement Description	The system shall send notifications for bookings, orders, and updates to users.
Priority	Medium
Reason	Keeps users informed about activities.
Measurement	Notifications are delivered successfully.
Requirement Status	Implemented

TABLE 5.4-7: FRQ07 – NOTIFICATIONS

Field	Description
Requirement ID	FR-07
Requirement Name	AI Stylist Assistant
Requirement Description	The system shall provide an AI assistant to suggest hairstyles and grooming tips.
Priority	Medium
Reason	Adds innovation and personalization to the app.
Measurement	Users receive relevant AI responses.
Requirement Status	Implemented

TABLE 5.4-8: FRQ08 – AI ASSISTANT

Field	Description
Requirement ID	FR-08
Requirement Name	Salon Owner Management
Requirement Description	The system shall allow salon owners to manage services, employees, bookings, and products.
Priority	High
Reason	Essential for business operations.
Measurement	Owners can manage all salon-related data successfully.
Requirement Status	Implemented

TABLE 5.4-9: FRQ09 – SALON OWNER MANAGEMENT

Field	Description
Requirement ID	FR-09
Requirement Name	Admin Control Panel
Requirement Description	The system shall allow admins to manage users, salons, products, and system operations.
Priority	High
Reason	Required for platform control and monitoring.
Measurement	Admin actions are successfully applied.
Requirement Status	Implemented

TABLE 5.4-10: FRQ10 – ADMIN MANAGEMENT

Field	Description
Requirement ID	FR-10
Requirement Name	Home Service Option
Requirement Description	The system shall allow customers to request home service when supported by salons.
Priority	Medium
Reason	Provides flexibility and convenience for users.
Measurement	Home service bookings are successfully processed.
Requirement Status	Implemented

Non-Functional Requirements:

TABLE 5.4-11: NFRQ01 – SYSTEM PERFORMANCE

Field	Description
Requirement ID	NFRQ01
Requirement Name	System Performance

Requirement Description	The system shall respond to user actions such as loading pages and booking requests within an acceptable time limit.
Priority	High
Reason	Ensures a smooth and fast user experience.
Measurement	System response time is less than 3 seconds for most operations.
Requirement Status	Implemented

TABLE 5.4-12: NFRQ02 – SYSTEM SECURITY

Field	Description
Requirement ID	NFRQ02
Requirement Name	System Security
Requirement Description	The system shall protect user data through secure authentication and data handling mechanisms.
Priority	High
Reason	Protects sensitive user information.
Measurement	No unauthorized access; data is securely stored and transmitted.
Requirement Status	Implemented

TABLE 5.4-13: NFRQ03 – USABILITY

Field	Description
Requirement ID	NFRQ03
Requirement Name	Usability
Requirement Description	The system shall provide an easy-to-use and intuitive interface for all users.
Priority	High
Reason	Improves user satisfaction.
Measurement	Users can complete tasks easily without confusion.
Requirement Status	Implemented

TABLE 5.4-14: NFRQ04 – SYSTEM AVAILABILITY

Field	Description
Requirement ID	NFRQ04
Requirement Name	System Availability
Requirement Description	The system shall be available at all times with minimal downtime.
Priority	High
Reason	Users need continuous access.
Measurement	System uptime is at least 99%.
Requirement Status	Implemented

TABLE 5.4-15: NFRQ05 – SCALABILITY

Field	Description
Requirement ID	NFRQ05
Requirement Name	Scalability
Requirement Description	The system shall handle increasing numbers of users without performance degradation.
Priority	Medium
Reason	Supports future expansion.
Measurement	Stable performance under high load.

Requirement Status	Implemented
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TABLE 5.4-16: NFRQ06 – COMPATIBILITY

Field	Description
Requirement ID	NFRQ06
Requirement Name	Compatibility
Requirement Description	The system shall run on multiple devices and operating systems.
Priority	Medium
Reason	Ensures accessibility across platforms.
Measurement	Works on Android, iOS, and web.
Requirement Status	Implemented

TABLE 5.4-17: NFRQ07 – MAINTAINABILITY

Field	Description
Requirement ID	NFRQ07
Requirement Name	Maintainability
Requirement Description	The system shall be easy to maintain and update.
Priority	Medium
Reason	Reduces maintenance effort.
Measurement	Updates can be applied without issues.
Requirement Status	Implemented

TABLE 5.4-18: NFRQ08 – LOCALIZATION

Field	Description
Requirement ID	NFRQ08
Requirement Name	Localization
Requirement Description	The system shall support multiple languages including English and Arabic.
Priority	Medium
Reason	meet local user needs.

Measurement	Users can switch languages successfully.
Requirement Status	Implemented

5.5 Persons

To understand the users of the Freshline application, five personas have been created. Customers include salon owners, salon employees including stylists and hairdressers, admin owners, and admin themselves. The personas will allow the system design to be done according to what the users expect to get from it. As a result of this assessment, the system will become organized and offer efficiency in services.

Persona 1: Customer

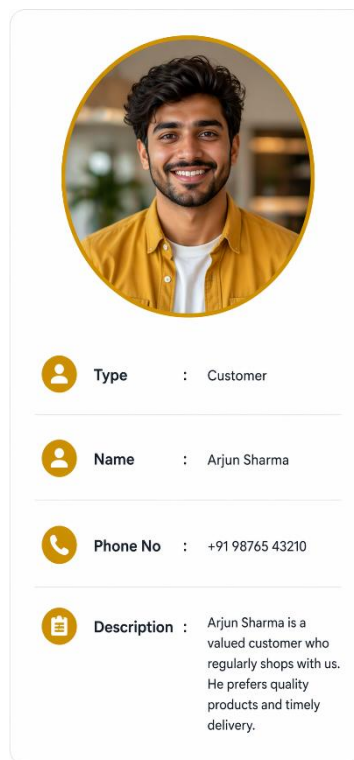


Figure 5.5-1. Persona Profile: Arjun Sharma

Arjun is the typical kind of customer who goes to salons to get his hair styled and buy beauty products. Sejal uses the Freshline platform to find salons that suit him, make instant reservations and save his time by skipping waiting lines. He appreciates simplicity, speed, transparency about prices and services, and convenience.

Persona 2: Salon Owner

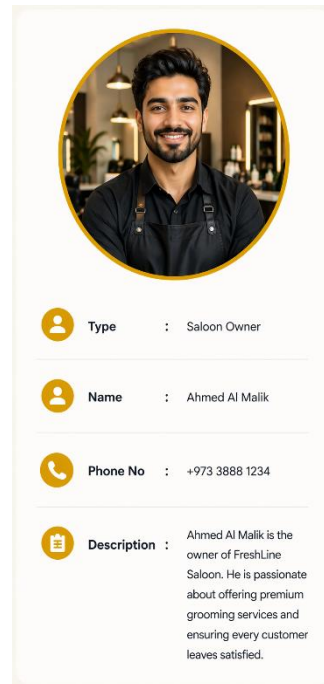


Figure 5.5-2. Persona Profile: Ahmed Al Malik

Ahmed is one of the owners of a salon who manages the day-to-day business operations and servicing of clients at his salon. One software creation used by Ahmed is Freshline to manage bookings, scheduling workers and organizing service activities at the salon. Ahmed likes a scheme to abolish unnecessary physical efforts and centralize everything.

Persona 3: Salon Employee

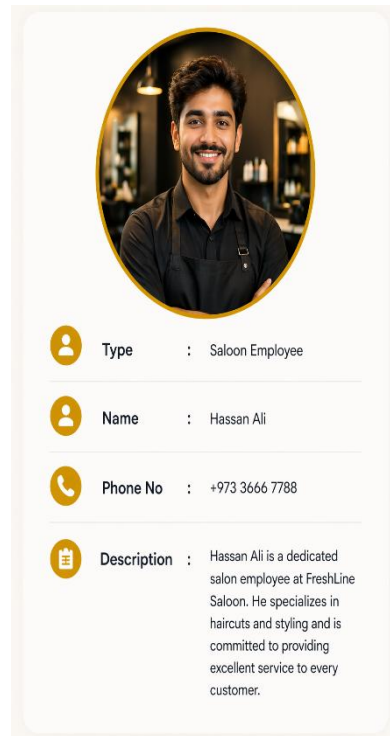


Figure 5.5-3. Persona Profile: Hassan Ali

Hassan acts as an intermediary between the employees and the salon providing grooming services. Hassan relies on the Freshline tools to tracks bookings and arrange his schedule. He enjoys organizing things methodically, which will help him schedule tasks. Hassan focuses on competency and quality service for its client.

Persona 4: Admin Owner

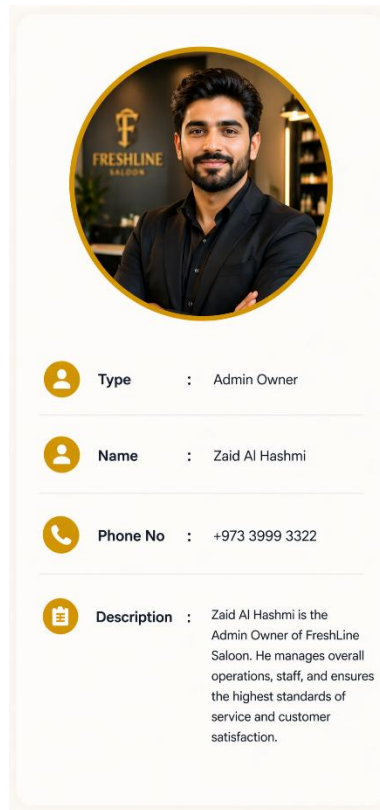


Figure 5.5-4. Persona Profile: Zaid Al Hashmi

Zaid, the admin owner, controls the operations of both the whole site and the salon. He manages the users, their activities, and the system's functionality through Freshline. He likes that the system is reliable; it has control options and security.

Persona 5: Admin

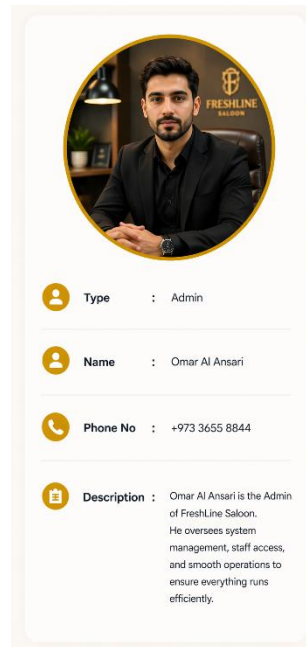


Figure 5.5-5. Persona Profile: Omar Al Ansari

Omar is a representation of the system administrator responsible for overseeing the day-to-day operations of the platform. He uses the Freshline system to manage access to the system, monitor what is done in this system, and to check the functionality. He is fond of orderly and stable systems.

5.6 System Models

The diagrams depicting the flow of information within the Freshline app gives an overall view of how information flows throughout the system, how user activity is controlled in the software and how the systems work. The Data Flow Diagrams (DFDs) of Level 0 and Level 1 for users like admin, customer and salon owner. The registration process, authentication, salon search, bookings, product purchase and services management will be analyzed. Process specifications refer to a detailed description of each process with respect to their inputs and outputs in the process. Also, the ERD shows how the data is organized in the database and how admin, salon, and customer are related.

5.6.1 Data Flow Diagram

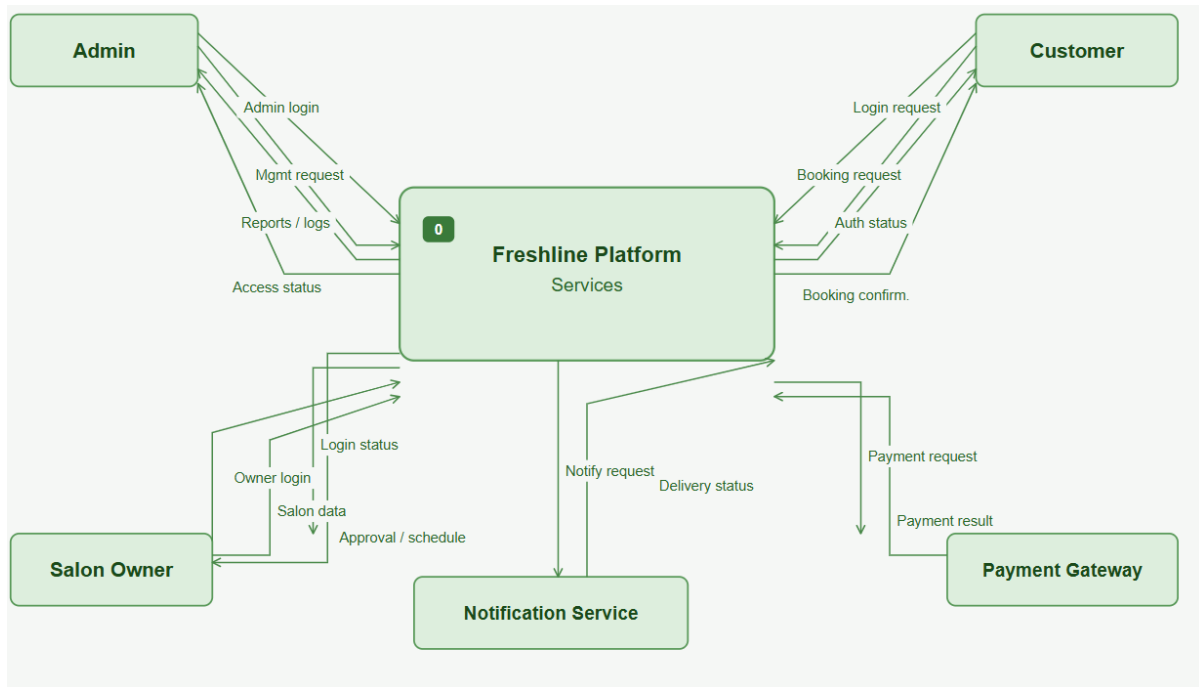


Figure 5.6-1. Data Flow Diagram Level 0

The DFD of Level 0 of Freshline is shown in this figure. It shows how all parties of the system are connected and perform any action in the system. Using login credentials, booking requests and booking confirmation messages, the customers interact with the system. Owners of the salons use the system for the management of salon-related information and the approval of the booking.

The user's activities are responded to by the system in the form of authentication confirmation success, booking confirmation and access to the user accounts. The system also communicates with the payment gateway to send a payment request and receive payment transaction results. Furthermore, the system can utilize the notification service for dispatching notifications.

In conclusion, the whole ecosystem is integrated on the Freshline platform, and each user and external service is communicating through a core system which ensures that all participants are communicating well and their requests are being processed accurately.

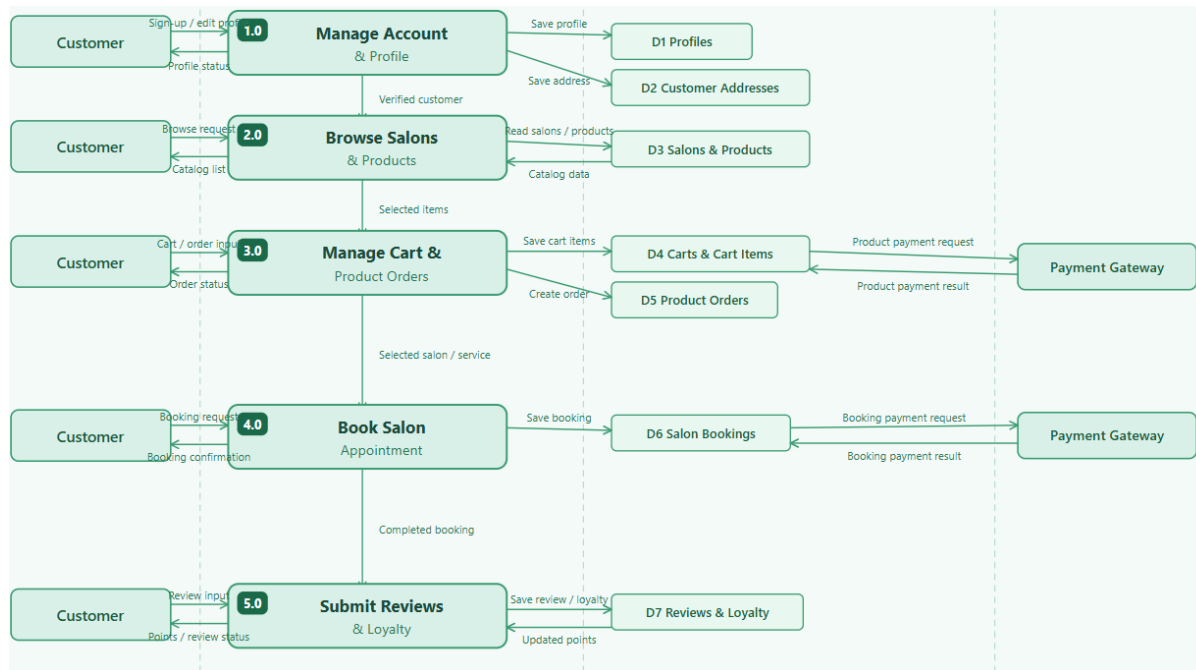


Figure 5.6-2. Data Flow Diagram Level 1 (Customer Module)

Below is figure 5.6.1-2, which represents the DFD Level 1 of the Customer Module in freshline. Operations of a customer are made up of five major processes. It includes Managing Account & Profile, Browsing Salons & Products, Managing Cart & Product Orders, Booking Salon Appointment, and Submitting Reviews & Loyalty. The first functionality is an account management system where a user will be able to register, login and administer their profile details. Once you have successfully accessed the account, you will be able to view various salons as well as their products. The next processing – manage cart & product orders, will let users add and order product from the app through the payment gateway. The Appointment Booking process helps the customers to make an appointment for any service. Users will submit reviews and earn loyalty points in the Submit Reviews & Loyalty process at last.

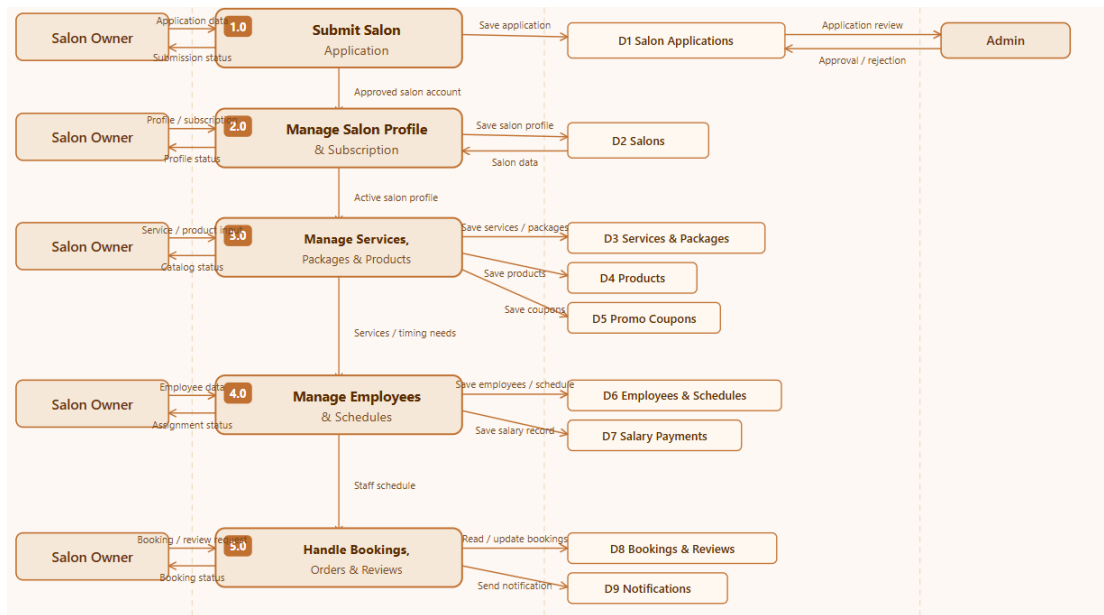


Figure 5.6-3. Data Flow Diagram Level 1 (Salon Owner Module)

The following image, Figure 5.6-3, represents the DFD Level 1 of the Freshline Salon Owner Module. The salon owner's operation is classified in five major processes, namely Submission of Salon Application, Management of Salon Profile and Subscription, Management of Services, Packages, and Products, Management of Employees and Schedules, and Management of Bookings, Orders, and Reviews. The procedure starts with filing the application in order to allow the salon to get approved or rejected by the admin. If granted approval, the owner of the salon will then be able to manage their profile as well as update the services offered and products. The owner is also able to manage subscriptions. In addition, the owner can control the information about the employees, assign them schedules, and provide them information for salaries. In addition, the owner can control bookings, orders and reviews.

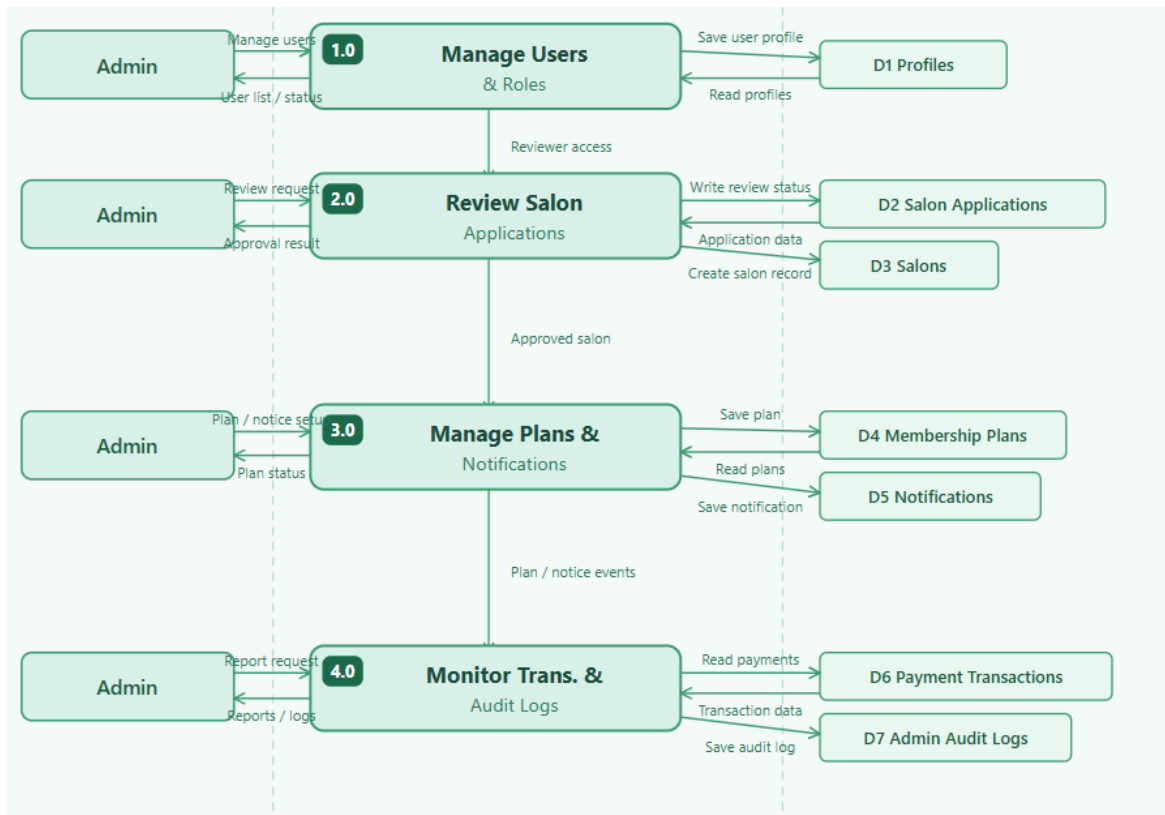


Figure 5.6-4. Data Flow Diagram Level 1 (Admin Module)

As shown in Figure 5.6-4, this is the DFD Level 1 for Freshline Admin Module. The end-user procedures primarily consist of 4 processes namely Manage Users & Roles, Review Salon Applications, Manage Plans & Notifications and Monitor Transactions & Audit Log. User management is the first process where the admin views the profiles, assign roles and controls access. After that, the admin assesses the salon applications, approves or rejects their submission and creates a salon account. With Process Manage Plans & Notifications, the admin can manage membership plans and inform users of the same. Finally, the last process which is the Monitor Transactions & Audit Logs lets the admin monitor payments and audit logs.

5.6.2 Process Specification

Level 1 for Freshline (Admin):

The Freshline platform admin processes are discussed in detail in this section. The process specification fully describes each Level-1 process and highlights the purpose of the process, the input, and the output. Level-1 processes include user management, role management, salon app assessment, plan and notifications management, and live transaction monitoring, among other things. There is a table given below which explains these.

TABLE 5.6.2-1: DFD LEVEL 1 - MANAGE USERS & ROLES

Field	Description
Process ID	1.0
Process Name	Manage Users & Roles
Process Description	Handles user accounts by creating, updating, assigning roles, and controlling access permissions.
Process Input	User data, role update requests
Process Output	Updated user profiles, role assignments, access status

TABLE 5.6.2-2: DFD LEVEL 1 – REVIEW SALON APPLICATIONS

Field	Description
Process ID	2.0
Process Name	Review Salon Applications
Process Description	Reviews submitted salon applications and approves or rejects them based on provided information.
Process Input	Salon application data, review request
Process Output	Approval or rejection result, created salon record

TABLE 5.6.2-3: DFD LEVEL 1 – MANAGE PLANS & NOTIFICATIONS

Field	Description
Process ID	3.0
Process Name	Manage Plans & Notifications
Process Description	Manages membership plans and sends notifications to users regarding updates and activities.
Process Input	Plan data, notification request
Process Output	Updated plans, sent notifications

TABLE 5.6.2-4: DFD LEVEL 1 – MONITOR TRANSACTIONS & AUDIT LOGS

Field	Description
Process ID	4.0
Process Name	Monitor Transactions & Audit Logs
Process Description	Tracks payment transactions and records system activities for monitoring and auditing purposes.
Process Input	Transaction data, report request
Process Output	Payment reports, audit logs, system activity records

Level 1 for Freshline (Customer):

Customer processes of Freshline platform will be explained in this section. Each level one process will be elaborated through process specification defining the objectives, inputs, and outputs for each process. Customer processes include Account management process, browse salon and products process, Cart, and orders management process, Booking appointments process, and Reviews submissions process. Processes and data flow of each process have been outlined in the tables below.

TABLE 5.6.2-5: DFD LEVEL 1 – MANAGE ACCOUNT & PROFILE

Field	Description
Process ID	1.0
Process Name	Manage Account & Profile
Process Description	Handles customer registration, login, and profile updates including personal information and addresses.
Process Input	User credentials, profile data
Process Output	Updated profile, authentication status

TABLE 5.6.2-6: DFD LEVEL 1 – BROWSE SALONS & PRODUCTS

Field	Description
Process ID	2.0
Process Name	Browse Salons & Products
Process Description	Allows customers to view salons, services, and products available on the platform.
Process Input	Browse request
Process Output	Salon listings, product catalog

TABLE 5.6.2-7: DFD LEVEL 1 – MANAGE CART & PRODUCT ORDERS

Field	Description
Process ID	3.0
Process Name	Manage Cart & Product Orders
Process Description	Handles adding products to cart, managing cart items, and placing product orders.
Process Input	Selected products, cart actions
Process Output	Updated cart, created product order

TABLE 5.6.2-8: DFD LEVEL 1 – BOOK SALON APPOINTMENT

Field	Description
Process ID	4.0
Process Name	Book Salon Appointment
Process Description	Allows customers to select services, choose time, and confirm salon bookings.
Process Input	Booking request, selected services
Process Output	Booking confirmation, saved appointment

TABLE 5.6.2-9: DFD LEVEL 1 – SUBMIT REVIEWS & LOYALTY

Field	Description
Process ID	5.0
Process Name	Submit Reviews & Loyalty
Process Description	Enables customers to submit reviews and earn or update loyalty points.
Process Input	Review data, rating input
Process Output	Stored review, updated loyalty points

Level 1 for Freshline (Salon Owner):

The processes of the Freshline application for salon owners are illustrated in this part. Every process in Level-1 is elaborated with process specifications, which describe the objective, input, and output of each process. The processes include applying for a salon, managing profiles, managing services and products, scheduling employees, and booking appointments. The following tables demonstrate the functioning of each process.

TABLE 5.6.2-10: DFD LEVEL 1 – SUBMIT SALON APPLICATION

Field	Description
Process ID	1.0
Process Name	Submit Salon Application
Process Description	Handles the submission of salon application details for admin review and approval.
Process Input	Application data, salon details
Process Output	Application status, approval or rejection result

TABLE 5.6.2-11: DFD LEVEL 1 – MANAGE SALON PROFILE & SUBSCRIPTION

Field	Description
Process ID	2.0
Process Name	Manage Salon Profile & Subscription
Process Description	Allows the salon owner to update profile details and manage subscription plans.
Process Input	Profile data, subscription updates
Process Output	Updated salon profile, subscription status

TABLE 5.6.2-12: DFD LEVEL 1 – MANAGE SERVICES, PACKAGES & PRODUCTS

Field	Description
Process ID	3.0
Process Name	Manage Services, Packages & Products
Process Description	Handles adding, updating, and removing services, packages, and products.
Process Input	Service data, product details
Process Output	Updated catalog, service and product listings

TABLE 5.6.2-13: DFD LEVEL 1 – MANAGE EMPLOYEES & SCHEDULES

Field	Description
Process ID	4.0
Process Name	Manage Employees & Schedules
Process Description	Manages employee records, schedules, and salary information.
Process Input	Employee data, schedule details
Process Output	Updated employee records, assigned schedules

TABLE 5.6.2-14: DFD LEVEL 1 – HANDLE BOOKINGS, ORDERS & REVIEWS

Field	Description
Process ID	5.0
Process Name	Handle Bookings, Orders & Reviews
Process Description	Allows the salon owner to manage bookings, track orders, and respond to customer reviews.
Process Input	Booking data, order details, review input
Process Output	Updated booking status, order tracking, managed reviews

5.6.3 – Entity relation Diagram

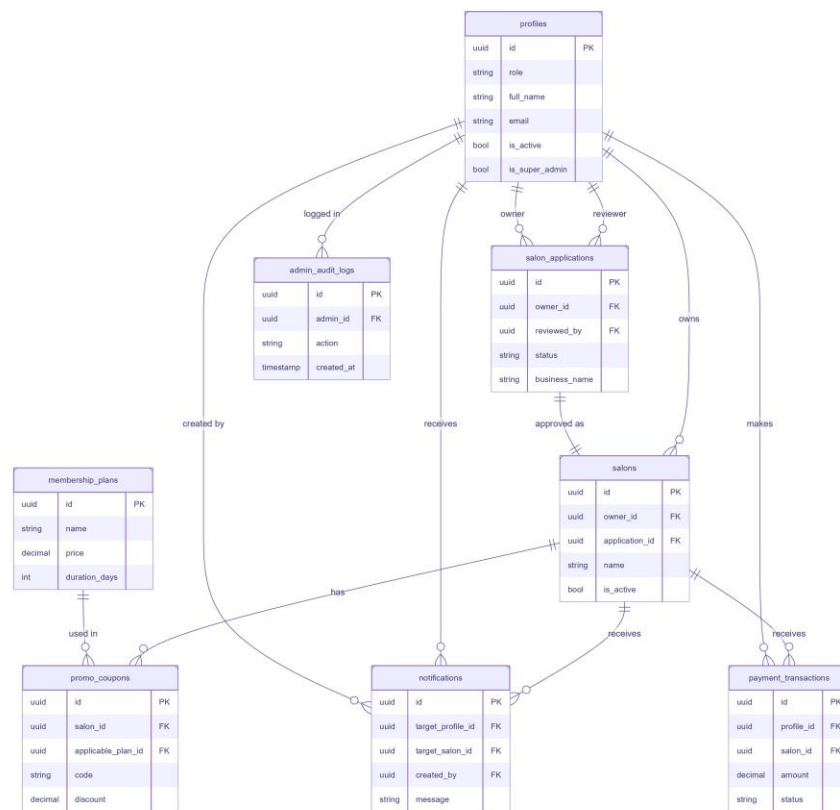


Figure 5.6.3-1. Entity Relationship Diagram (Admin Module)

From Figure 5.6.3-1, the ERD represents the interaction between the admin and various entities in the Freshline platform. Profiles are the primary entities used in the database and link other entities such as salon_applications and salons to capture, review, and approve applications for salon profiles. The admin also oversees system activities through the admin_audit_logs that capture any system activity and update details. Subscriptions and promotions are managed using membership_plans and promo_coupons, respectively, while system notifications are sent to users through the notifications entity. Payment_transactions made as part of the system activities are stored under payment transactions.



Figure 5.6.3-2. Entity Relationship Diagram (Customer Module)

The ERD in Figure 5.6.3-2 shows how he/she interacts with the main components of the Freshline platform. The table of profiles serves as the core table of the database to record information regarding the customer. Further, it links to relevant entities like customer_addresses, saved_cards, and the loyalty_wallets. Overall, these entities manage one's personal information and customer preferences. The construction follows the same pattern as orders where cart and cart_items are used to select the product/product-array. Likewise, product_order and product_order_items are also implemented for selective ordering. Customers can use the salons and salon_bookings entities to browse and book salons. Moreover, feedback from product_reviews and salon_reviews help customers, while notifications help communicate alerts and updates. All customer interactions like browsing, booking, buying, reviewing etc. are managed by these connections in a structured and user-friendly manner.

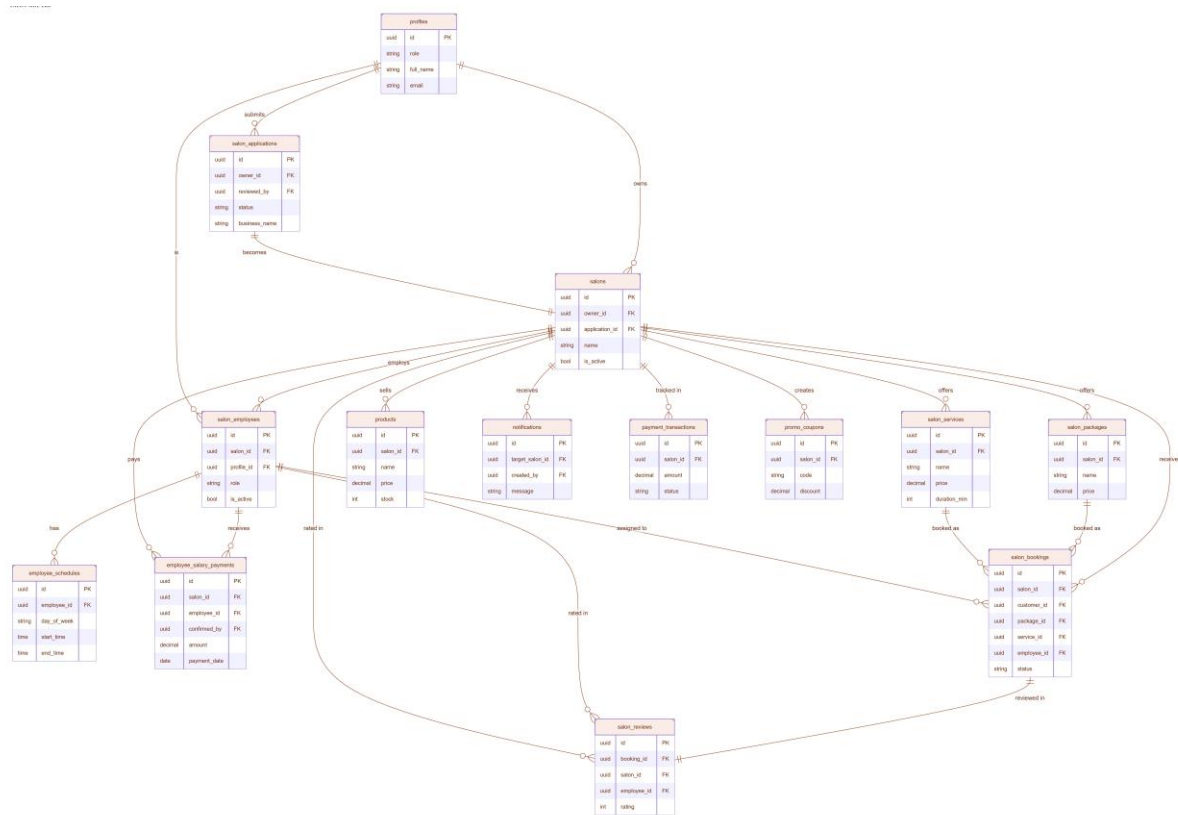


Figure 5.6.3-3. Entity Relationship Diagram (Salon Owner Module)

According to Figure 5.6.3-3, this ERD shows how the salon owner interacts with the main components of the Freshline platform. The profiles table acts as the central entity, linking to salons and salon_applications, which handle the submission and approval of salon accounts. The salon entity connects to related entities such as salon_employees, employee_schedules, and employee_salary_payments, which manage staff and operations. The system also includes products, salon_services, and salon_packages to manage service and product offerings. The salon_bookings entity handles customer appointments, while salon_reviews store feedback from customers. In addition, promo_coupons, notifications, and payment_transactions manage promotions, communication, and financial activities. These connected entities ensure that all salon operations are organized and efficiently managed, providing a structured and centralized system for business management on the Freshline platform.

Chapter 6

System Design

6.1 Introduction

In this chapter, the design of the proposed Freshline system will be explained. The design process will take care of creating a system that will have features as per the system requirement. In the design process, the system requirements will be translated into a system design. The design process will explain how the different elements in the system interact with each other and how they work.

This chapter will focus on important issues regarding the design of the system. System architecture, database design, and user interface design are the most important issues related to design. Moreover, this chapter will also discuss the design of the system that will support multiple users like customers, salon owners, employees, and administrators. Finally, this chapter will highlight some best practices that must be followed while designing the system.

6.2 Splash screen

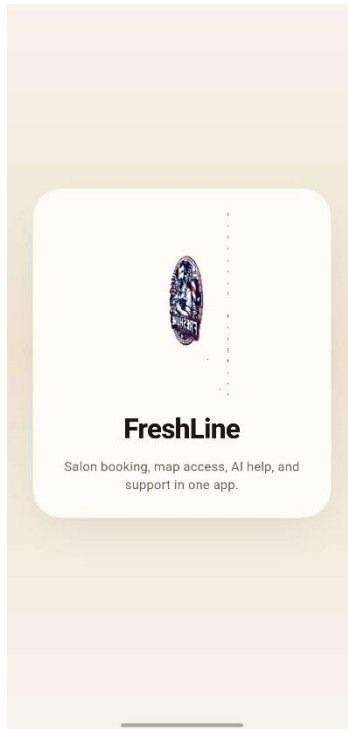


Figure 6.2- 1. Splash Screen

The splash screen shows the logo and title of the Freshline program, as well as a small description of what the application does. See Figure 6.2-1 below. The splash screen acts as an initial visual contact that users have with the application once the program is run. This screen not only helps to create the brand image of the application, but it also acts as a way to introduce the application to users.

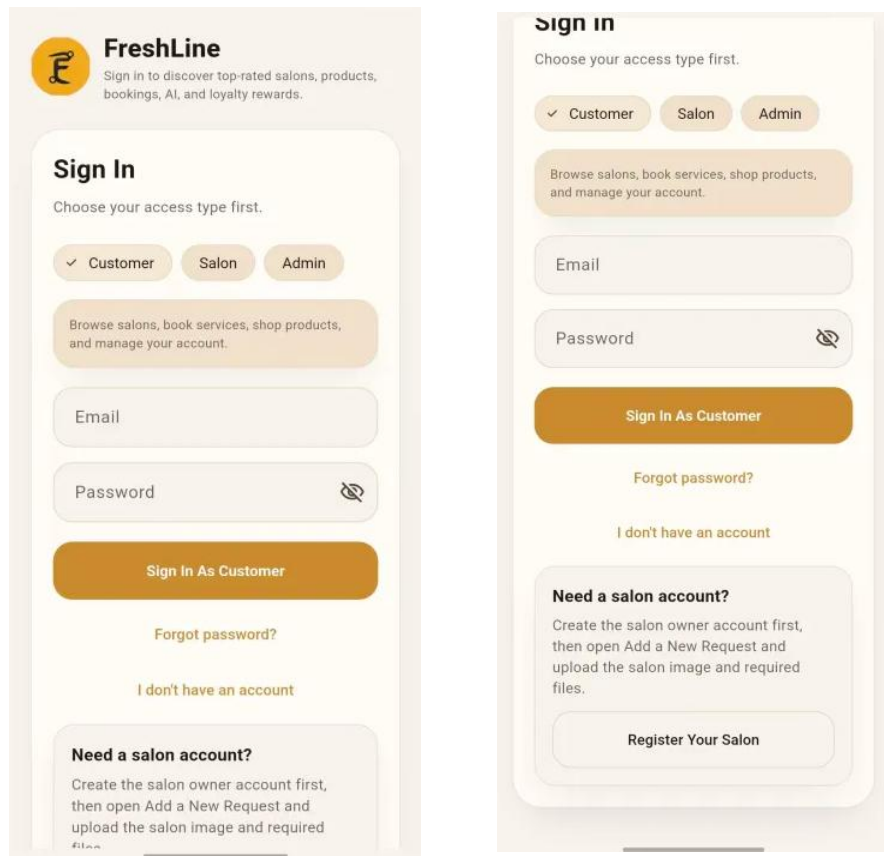
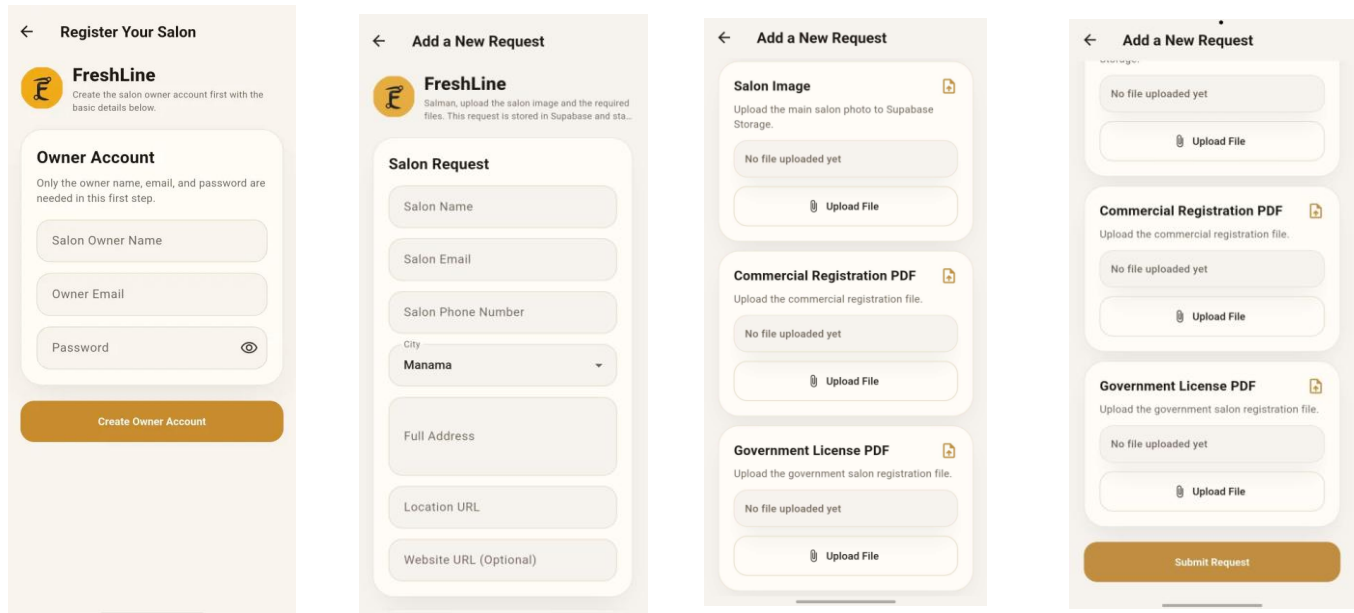


Figure 6.2- 2. Sign screen

The sign-in page is where the user can sign in to the Freshline app using his credentials, as illustrated in Figure 6.2-2. The sign-in page has many access levels, such as customer, salon, and administrator, giving various kinds of users access to their preferred accounts within the software. There are email address and password input boxes, as well as controls for viewing the password and resetting the password, on the sign-in interface. The sign-in interface also offers navigation links for creating a new account and registering the salon for its owner.

6.3 Salon Role



The figure displays four sequential mobile app screens for salon registration:

- Register Your Salon:** Features the FreshLine logo and a prompt to create an owner account. It includes input fields for "Salon Owner Name", "Owner Email", and "Password" (with a toggle for visibility). A "Create Owner Account" button is at the bottom.
- Add a New Request (Salon Request):** Prompts the user to upload a salon image and required files. It contains input fields for "Salon Name", "Salon Email", "Salon Phone Number", "City" (with a dropdown menu showing "Manama"), "Full Address", "Location URL", and "Website URL (Optional)".
- Add a New Request (Salon Image):** Specifically for uploading the "Salon Image". It includes a text prompt, a "No file uploaded yet" status, and an "Upload File" button.
- Add a New Request (Document Uploads):** For uploading "Commercial Registration PDF" and "Government License PDF". Each document section has a text prompt, a "No file uploaded yet" status, and an "Upload File" button. A "Submit Request" button is at the bottom.

Figure 6.3- 1. Registration for salon

The screens for salon registration enable salon owners to register for an account on Freshline and submit a request for joining, as illustrated in Figure 6.3-1 below. This is done to guarantee the inclusion of salons in the platform that have been verified.

The initial step involved in the process is registering an owner account, and it entails filling the mandatory fields such as the owner's name, email address, and password. Following this, the salon owner proceeds to fill in the other details regarding the salon. These include salon name, contact information, location, and salon address. The process also involves submission of necessary documents such as the image of the salon, business registration, and government license. This approach makes sure that all relevant information is collected in an organized way.

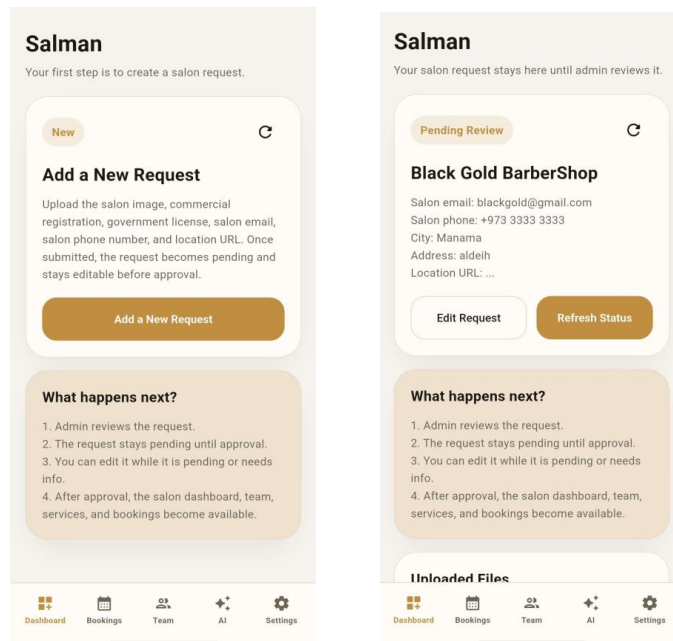


Figure 6.3- 2. Salon Dashboard Screen

The salon dashboard screen is provided in the context of an overview of request status and potential actions of the salon owner in the Freshline system and is represented in Figure 6.3-2 below. This screen is different depending on the fact whether the request was already sent by the salon.

Prior to sending the request, the salon owner has access to the button of sending a new request, as well as the information about what data is necessary and how the approval process looks like. Once the request is already sent, then this screen reflects all data related to it, namely the salon data and its request status, such as pending for approval. Moreover, there are buttons to edit the request and update the status.

Finally, there is also a 'what happens next?' window with the description of further actions.

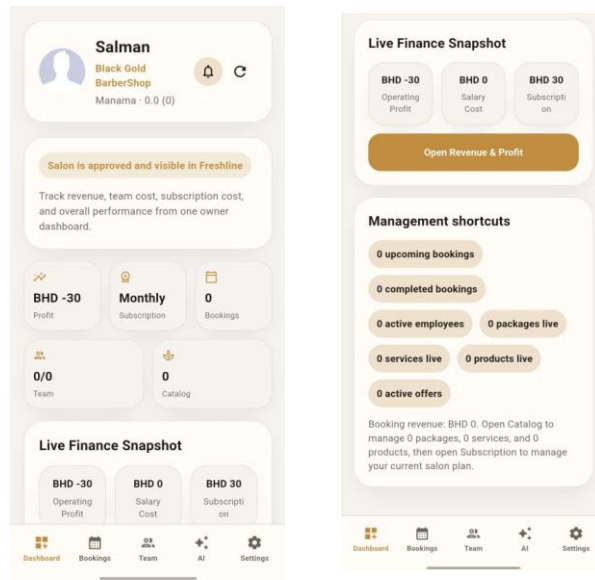


Figure 6.3- 3. Saloon Owner Dashboard

The salon dashboard screen in an approved state offers salon owners a dashboard that gives an overall look at their salon operations and management options, as seen in Figure 6.3-3. The salon dashboard screen automatically opens up when the salon owner's request is approved, showing critical information about the salon.

The dashboard includes performance information such as operational profit, staff salary, and subscription costs, among others. There are other booking, staff, and catalog indicators on the dashboard as well. The management options of the salon are provided by shortcut links to allow easy management of bookings, staff members, services, and products.

Lastly, the dashboard also shows the active state of the salon in the system, meaning that the salon is visible to all potential customers.

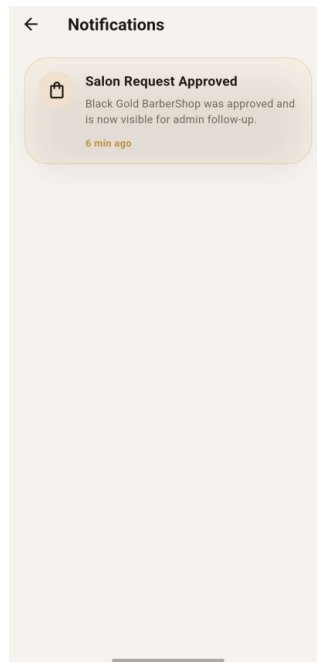


Figure 6.3- 4. Saloon Notification

The notifications window provides updates and system messages to the user, as depicted in Figure 6.3-4 below. This allows the user to be updated on various happenings on the system in the Freshline application, including notification of approval of the salon requests.

In this case, the notification provides an update to the salon owner informing him/her of the approval of his/her request by the system, which will be followed up by the admin. Each notification comes with a message together with the date of receipt of the message.

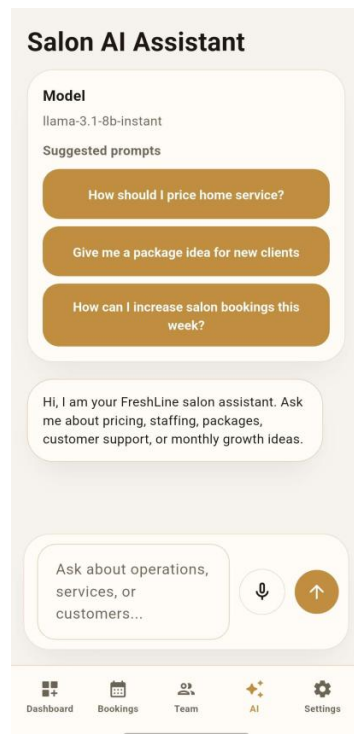


Figure 6.3- 5. Saloon owner ai screen

Figure 6.3-5 illustrates how the AI assistant interface lets the user communicate with the intelligent assistant for advice related to the salon. Users will be provided with possible prompt suggestions and an option to initiate conversations by asking questions about cost, service availability, and business management.

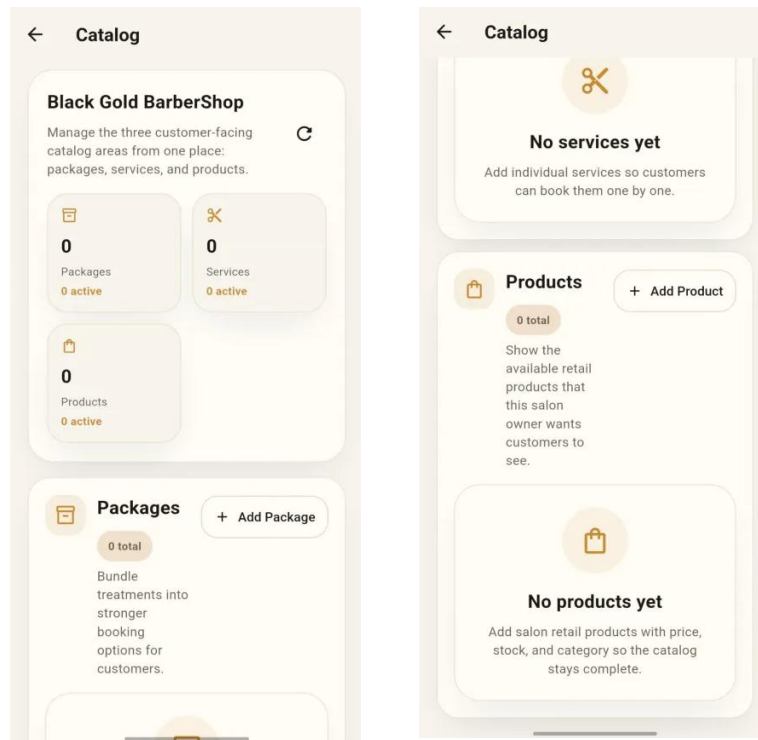


Figure 6.3- 6. CatLog Saloon owner screen

Figure 6.3-6 is the salon catalog page that enables the salon owner to administer his services, package and products. This page helps him create, arrange, and customize the entries within the catalog to be able to determine which services will be seen by customers.

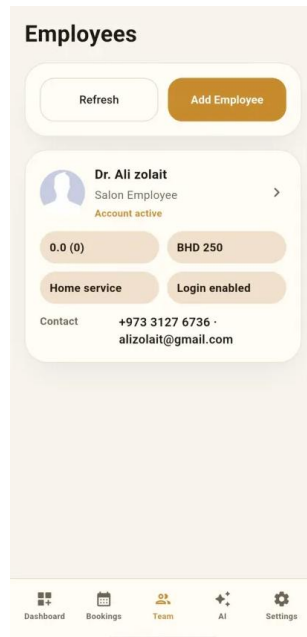


Figure 6.3- 7. Team saloon owner screen

The salon staff screen is what will enable the salon owner to control his or her staff, as seen in Figure 6.3-7. The screen shows the staff details including the job title, staff status, salaries, and contact information. It also gives room for adding new staff or modifying current staff details.

Add Employee

Employee profile image
Optional. If you skip this, Freshline will use the default profile image.

Employee name
Dr. Ali zolait

Role title
Salon Employee

Email
alizolait@gmail.com

Phone
+973 3127 6736

Monthly salary (BHD)
250

Password (optional)
.....

Supports in-shop bookings ☒

Add Employee

The entered password will be saved for the employee account.

Figure 6.3- 8. Add employee

The add employee screen enables the creation of employee profiles by salon managers, as illustrated in Figure 6.3-8. This screen features sections for adding the personal and professional data of employees, including their names, positions, contact numbers, salaries, and any desired login credentials. Other fields on the same screen allow the manager to define the abilities of each employee in relation to booking of in-store appointments, at-home services, and logging into the system.

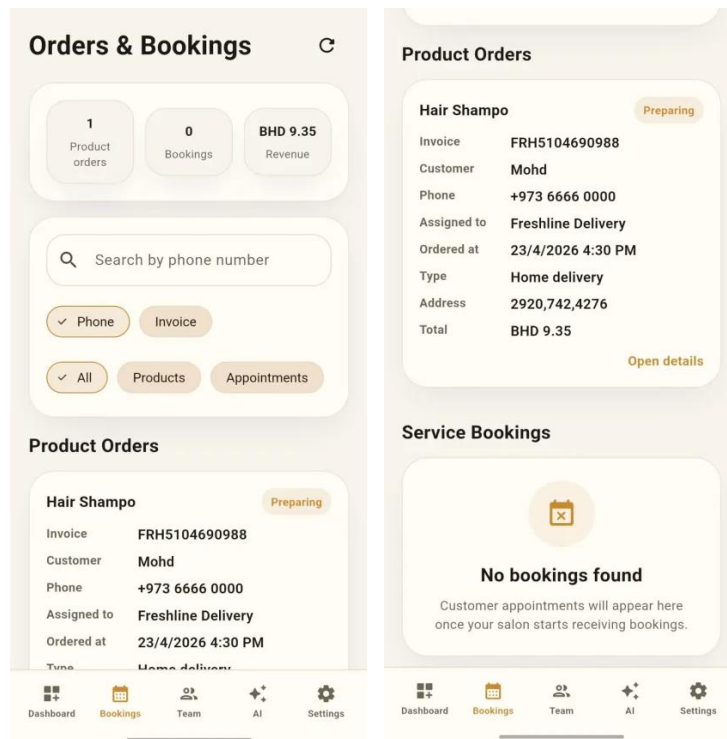


Figure 6.3- 9. Bookings

Product orders and appointments for services can be seen and managed through the Booking window, as illustrated in Figure 6.3-9 below. The Booking window provides information on orders, such as customer details, order status, and delivery mode. Filters are also provided to facilitate searching and sorting of records.

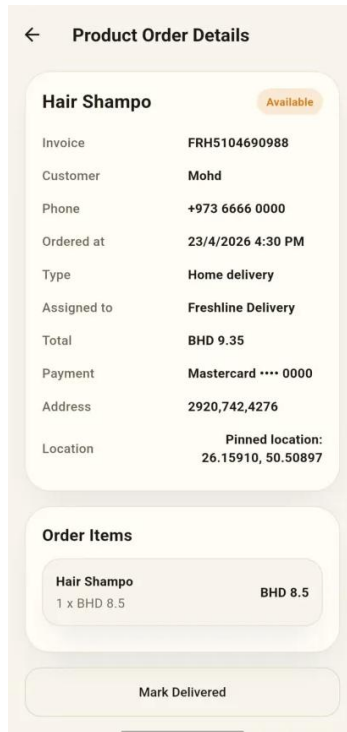


Figure 6.3- 10. Order Details

Details of order screen provides complete information regarding an order made by customers on any particular product, as illustrated in Figure 6.3-10. The screen presents the complete details of customers, order status, mode of delivery, payment mode, and location together with the ordered product.

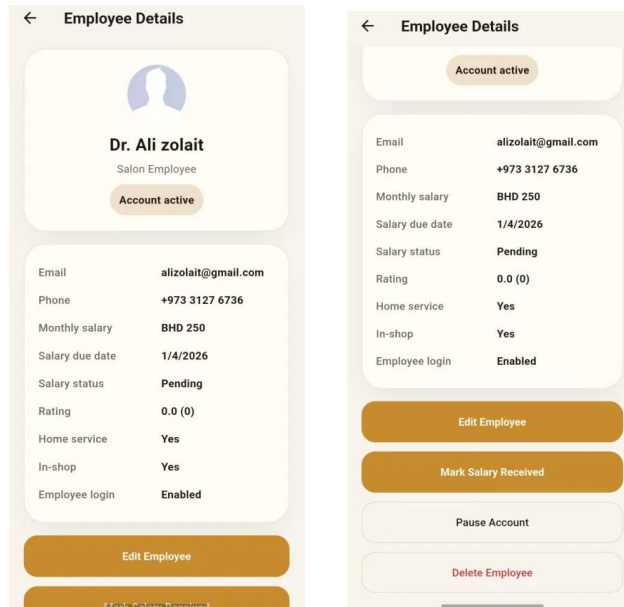


Figure 6.3- 11. Employee Details

Figure 6.3-11 presents the employee details screen which shows information regarding the selected employee. This screen provides both personal and professional information like contacts, salary, status, and capabilities. The user can perform various actions on the selected employee like editing information or changing the status of their salary.

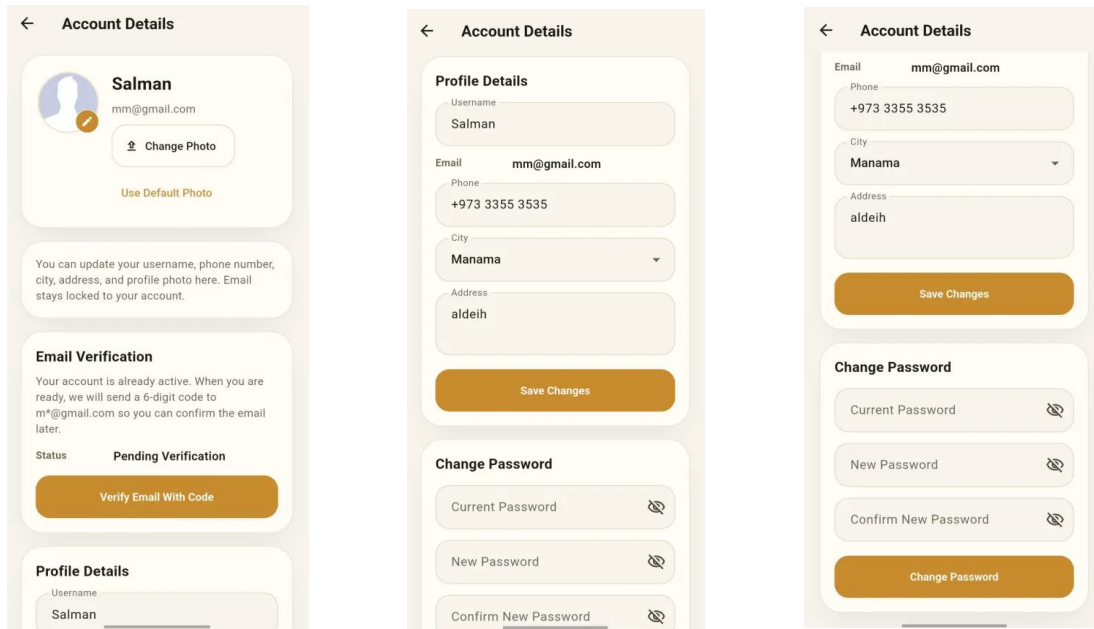


Figure 6.3- 12. Account Details

This is where the user can view and edit their personal data as illustrated in Figure 6.3-12. This section contains information regarding the profile, which includes the user's name, contact details, and address, among others. Users can also edit their profile picture and reset their password.

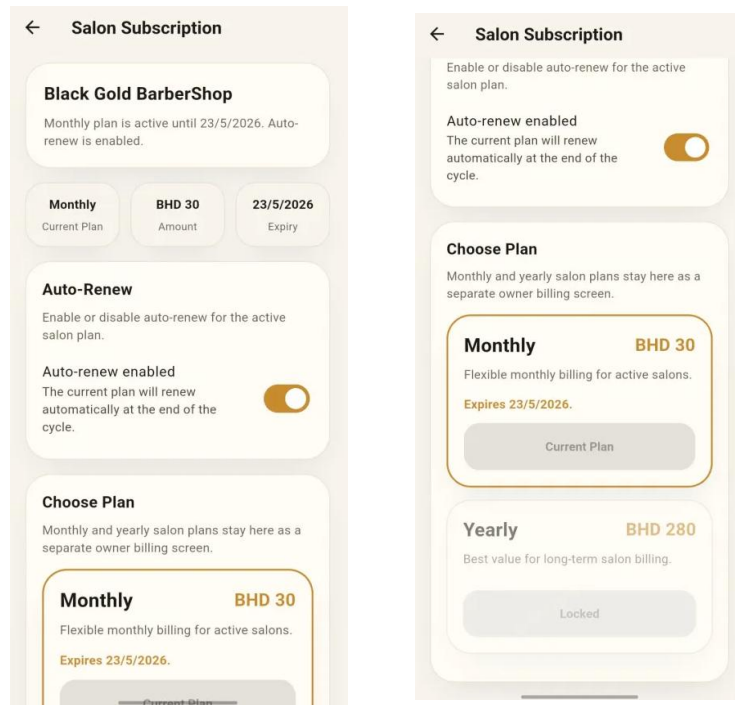


Figure 6.3- 13. Salon subscription

Figure 6.3-13 is an illustration of the salon subscription screen that shows salon owners how to see the subscription details. The screen features information about the existing plan, price, expiry date, auto-renewal option and other subscription plan choices, including activation and deactivation of the auto-renewal option.

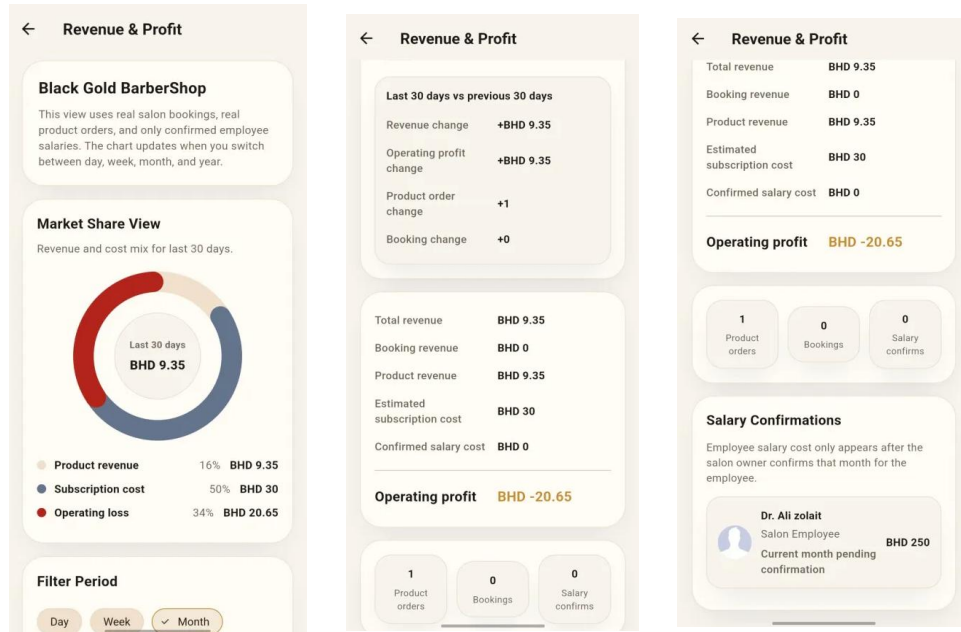


Figure 6.3- 14. Revenue & Profit

Salon owners can use the income and profit report screen to gauge the performance of their business operations, as depicted in Figure 6.3-14 below. This screen provides information on income, operating profit, and expenses, including graphical representations that allow salon owners to evaluate the performance of their business operations.

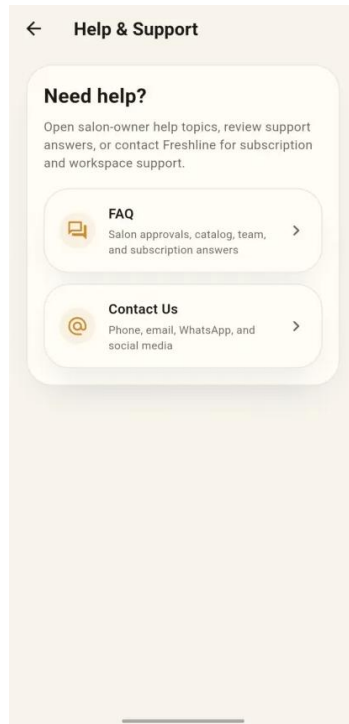


Figure 6.3- 15. Help & Support

The help and support screen enables the user to have access to information regarding help and support services, as illustrated in Figure 6.3-15 below. The help and support screen contains areas that include frequently asked questions and contact information.

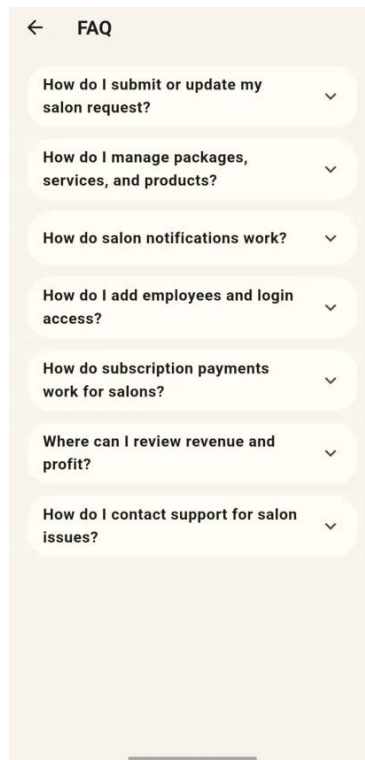


Figure 6.3- 16. FAQ

The FAQ screen enables the user to access a list of frequently asked questions and their respective answers, as illustrated in Figure 6.3-16. Users do not have to contact support directly for answers on how to use functionalities such as requests, services, notifications, and subscriptions.

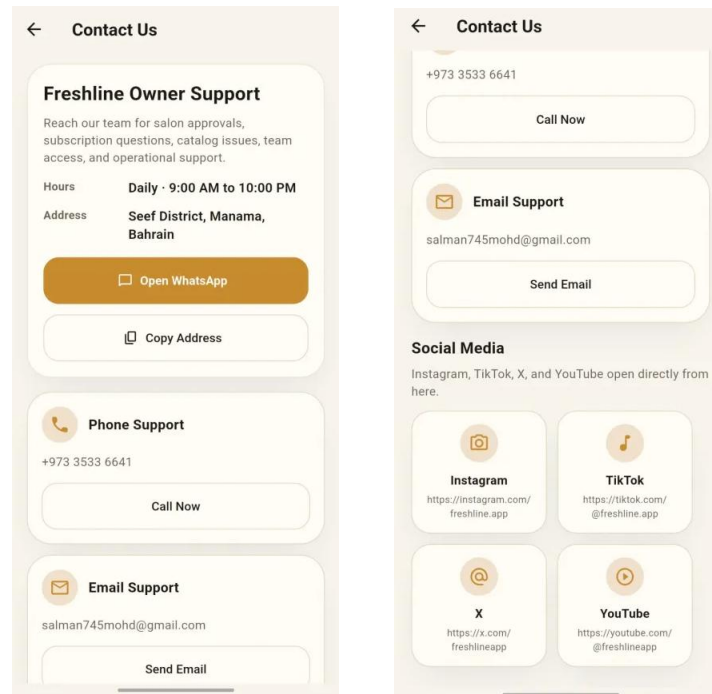


Figure 6.3- 17. Contact Us

The screen contacts us offers various methods of contacting customer support through a range of means, as depicted in Figure 6.3-17. The methods include telephone number, email address, WhatsApp link, and even social media connections.

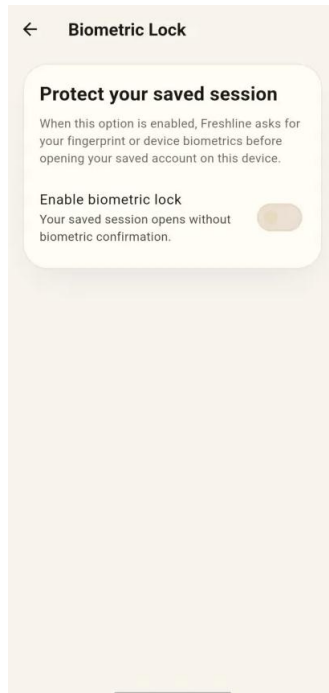


Figure 6.3- 18. Biometric screen

The biometric lock screen is used by users to allow or block access using biometrics, as illustrated in Figure 6.3-18. This screen can be used to access devices using biometrics like fingerprints in order to secure their saved session.

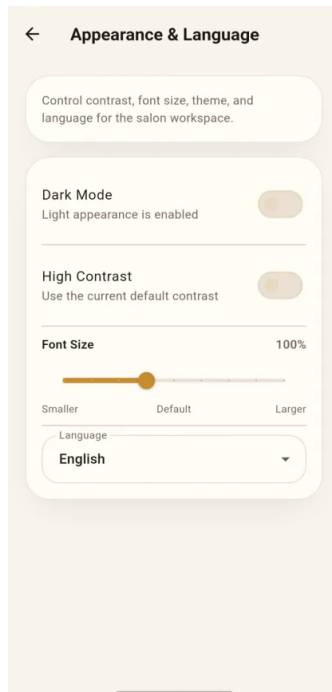


Figure 6.3- 19. Appearance & Language

This is where display settings can be customized using the appearance and language screen, as illustrated in Figure 6.3-19 below. The choices include dark mode, high contrast, font size changes, and language settings. Users will have an opportunity to adjust the application according to their liking.

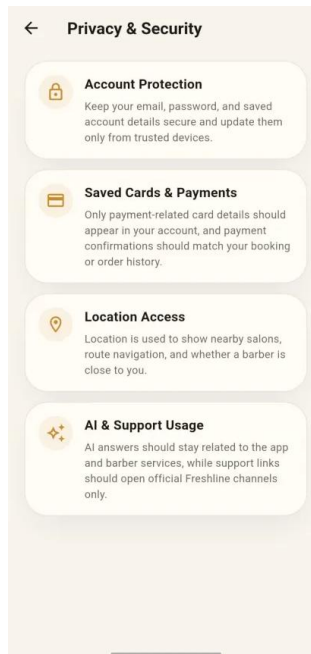


Figure 6.3-20. Privacy & Security

The privacy and security screen are where users get information regarding account security and security features of the app. As seen in Figure 6.3-20, some of the aspects addressed include account security, handling of payment information, location use, and artificial intelligence use.

6.4 Employee Role

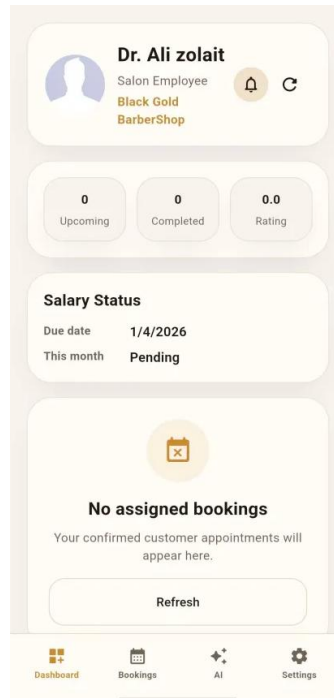


Figure 6.4-1. Employee Dashboard

The employee dashboard screen is used to provide the employees with a summary of their activity and tasks that have been assigned to them, as illustrated in Figure 6.4-1 below. The screen contains basic information like booking status, rating, salary, and assigned bookings.

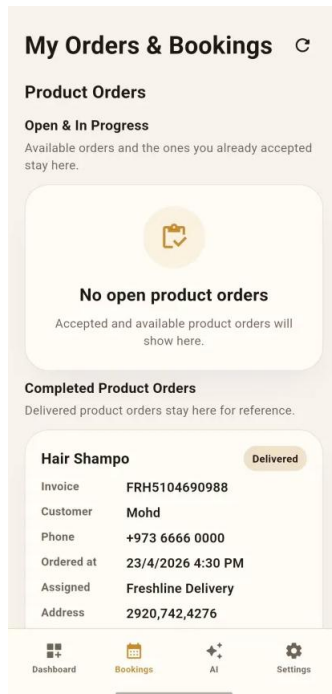


Figure 6.4-2. My Order & Bookings

The orders and bookings menu enables employees to manage orders and bookings. This menu is shown in Figure 6.4-2. The orders and bookings menu provides information about existing orders and bookings and allows employees to follow up on their orders.

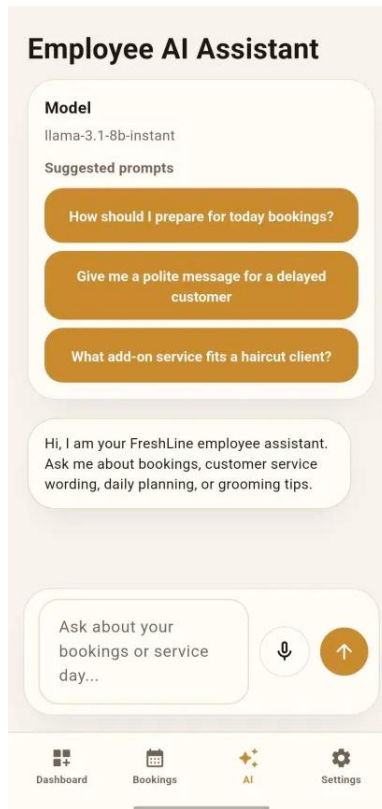


Figure 6.4-3. Employee Ai

This is the employee AI assistant screen which enables employees to communicate with the intelligent assistant for assistance and advice as illustrated in Figure 6.4-3 below. This gives prompts on what the employee can inquire about and even provides a chat box where employees can ask about bookings and general issues.

6.5 Admin Role

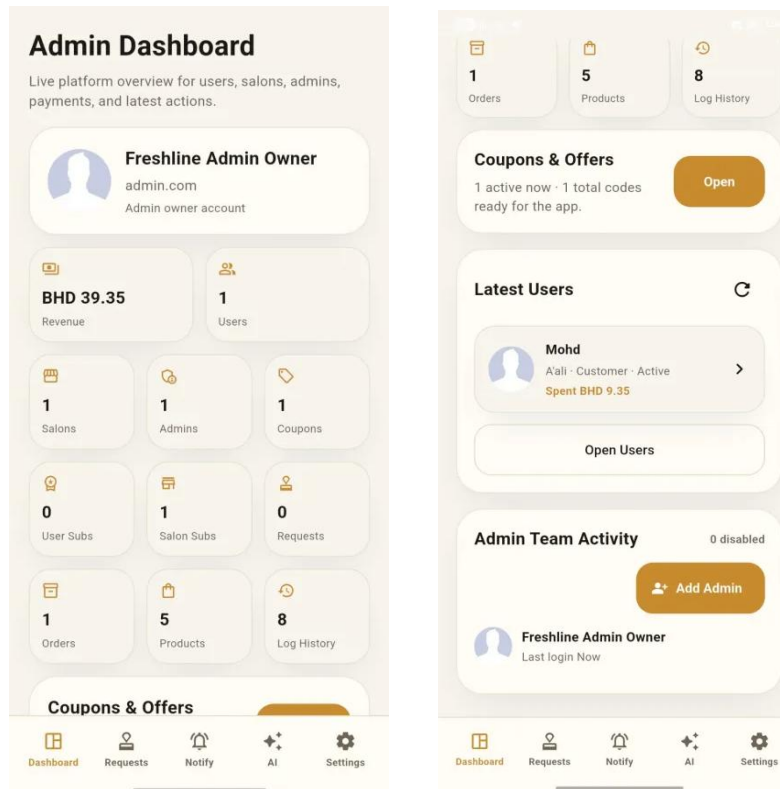


Figure 6.5-1. Admin Dashboard

The dashboard admin screen offers admins a complete look at everything that is going on within the system, as illustrated in Figure 6.5-1. This includes details like revenue, number of users, salons, subscriptions, orders, and system logs. It also includes tabs for coupon management, recent users, and admin activity.

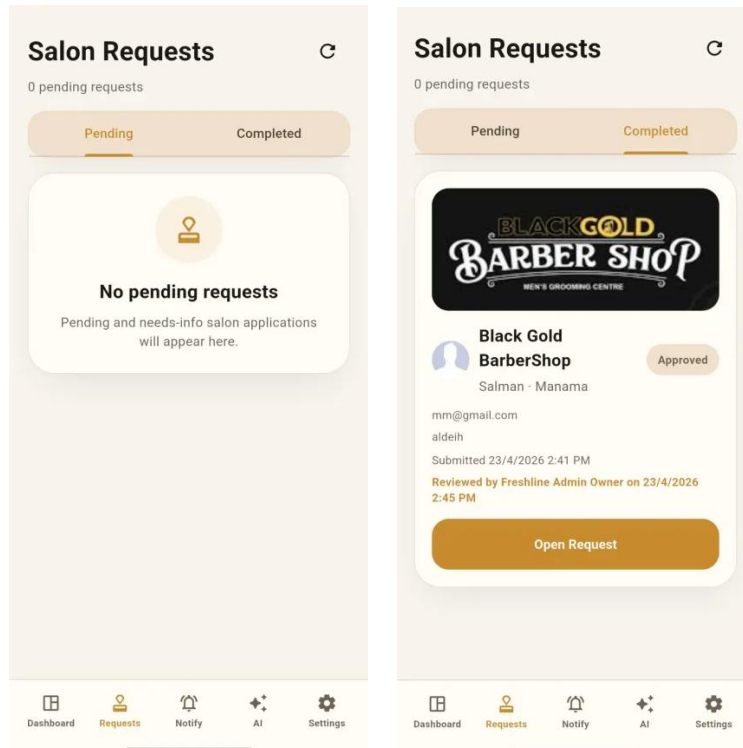


Figure 6.5-2. Salon Request

The Salon Requests screen allows managers to view and control salon request application. As illustrated in figure 6.5-2 below. The screen indicates all the requests that have been applied for and approved by the owner along with the full details of the name of the salon, owner's name, date of application and whether the approval has been given or not.

Notifications

Send platform message
Choose who should receive the notification, then send it to their in-app inbox.

✓ All Users Single User

All Salons Single Salon

Every customer account will receive this notification.

Notification Title

Message

➤ Send Notification

Dashboard Requests **Notify** AI Settings

Figure 6.5-3. Notification

The notification management page allows for sending notifications to the users within the platform, as depicted in Figure 6.5-3 below. It enables the administrator to select the recipient group, whether all users, a specific user, salon, or specific salons, while filling the notification title and message fields.

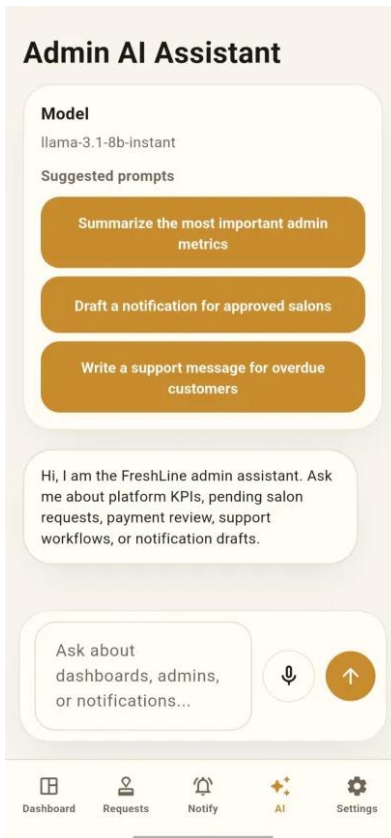


Figure 6.5-4. Admin Ai

The Administrator AI Assistant screen is where the administrator interacts with an intelligent assistant that helps with system administration tasks. This is illustrated in Figure 6.5-4. The administrator can get suggestions and use chat features to analyze metrics, process requests, and write notifications.

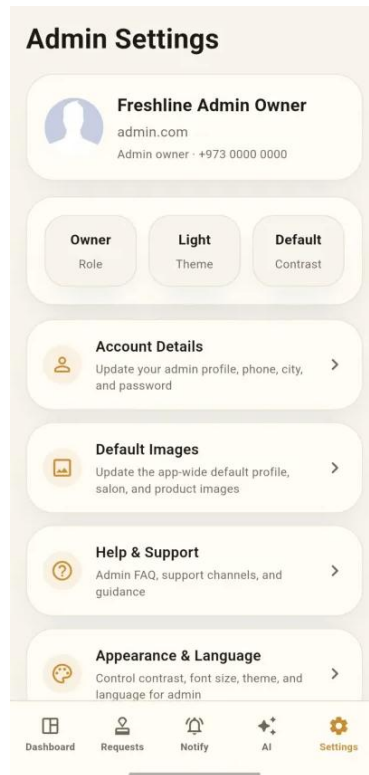


Figure 6.5-5. Admin Settings

Administrators have access to admin settings from the admin settings page where they can configure system settings and accounts settings as illustrated in Figure 6.5-5 below. The administrator can change account information, configure default images, seek help, customize appearance, and configure language settings.

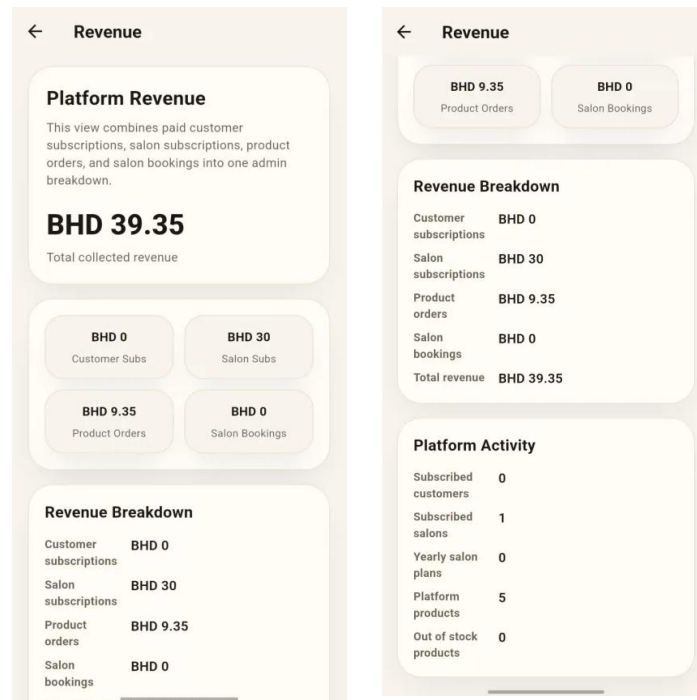


Figure 6.5-6. Revenue

The revenue page helps the administrator track the performance of the platform, which is demonstrated in Figure 6.5-6 below. It shows the total revenue generated from the subscription, product, and booking breakdowns, together with the platform's performance statistics.

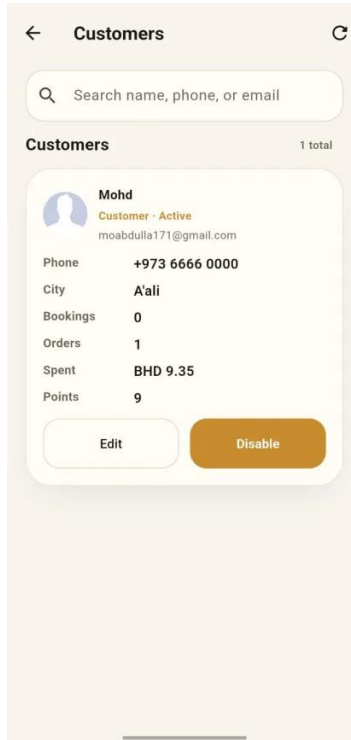


Figure 6.5-7. customer

Through the customer page, the administrator will be able to review or manage the customer accounts, as depicted in Figure 6.5-7 below. The page shows details about the customer, including their contact information, status, reservations, purchases, and expenditure.

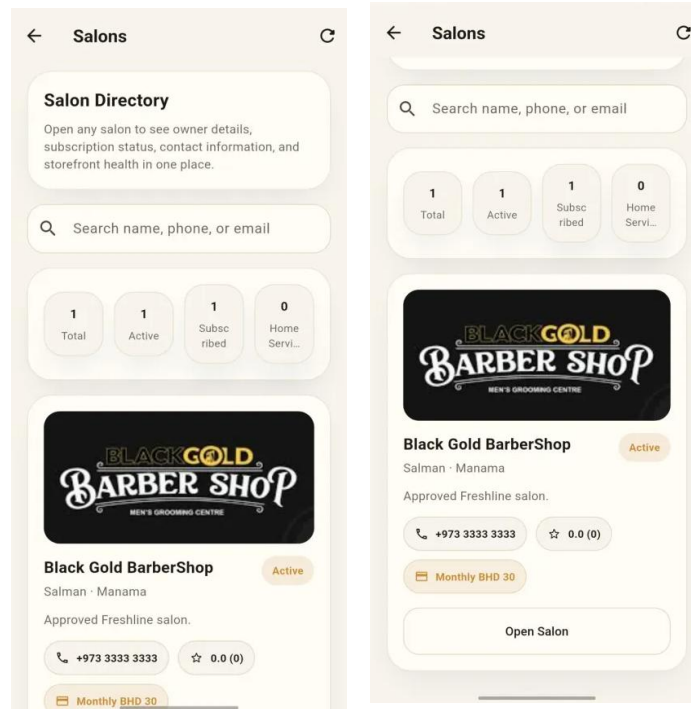


Figure 6.5-8. Salon details

The Salons screen enables the administrator to manage all the salons which have been registered as seen from Figure 6.5-8. The details which are displayed here include those about the salon name, location, status, and subscription, among others.

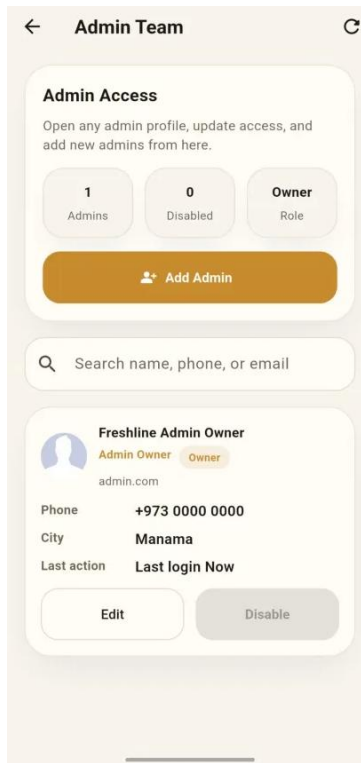


Figure 6.5-9. Admin team

Admin Team Screen is a window that provides administrative users the ability to administer their admin accounts and rights, as illustrated in Figure 6.5-9. The screen contains information about admin accounts in terms of role, contact details, and status. It also enables the addition, modification, or disabling of admin accounts.

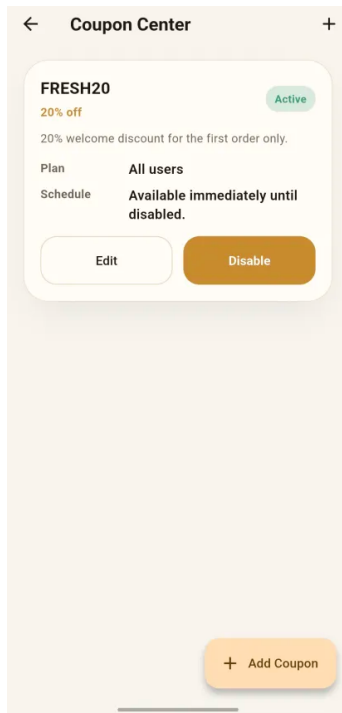


Figure 6.5-10. Coupon center

The coupon center form shows how you can use it to manage your coupons; it is depicted in Figure 6.5-10 below. The form gives you the opportunity to view the details about your coupons, including the discount percentage, validity status, availability, and target audience.

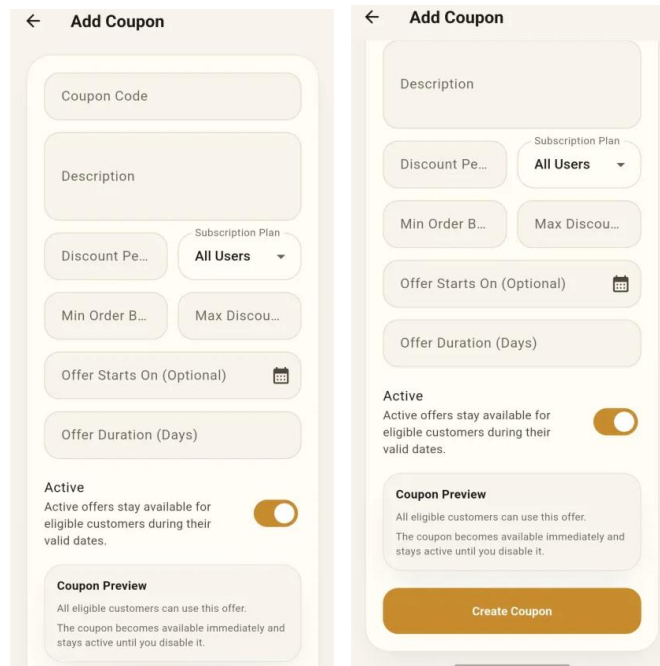


Figure 6.5-11. Add Coupon

Coupon addition screen helps in the creation of new coupon codes, as depicted in Figure 6.5-11 below. This enables easy administration by providing spaces for inputting information on the code number, description, discount percentage, conditions for application, period, and status.

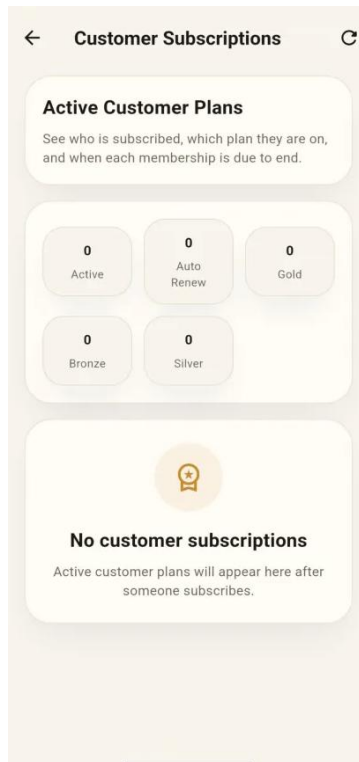


Figure 6.5-12. customer subscription

The customer subscriptions window enables the admin to track the customers' plans that are currently being used, as illustrated in Figure 6.5-12. This window shows subscription types, statuses, and activities.

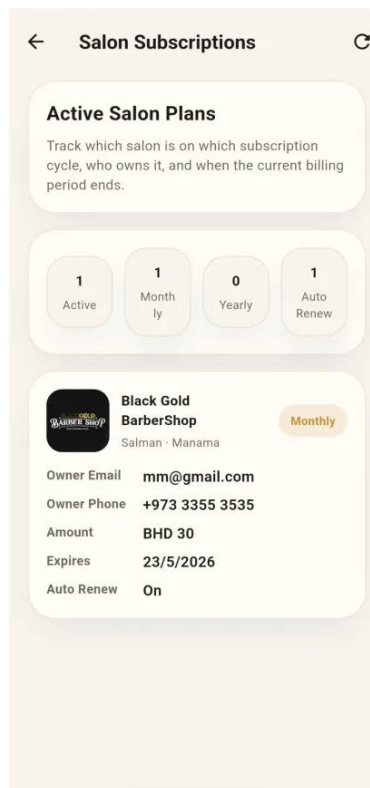


Figure 6.5-13. salon subscription

The page for viewing salon subscriptions shows the admin everything about salon subscriptions, as demonstrated in Figure 6.5-13. The screen displays information such as the subscription type, subscription costs, end dates, and whether the subscription auto-renews.

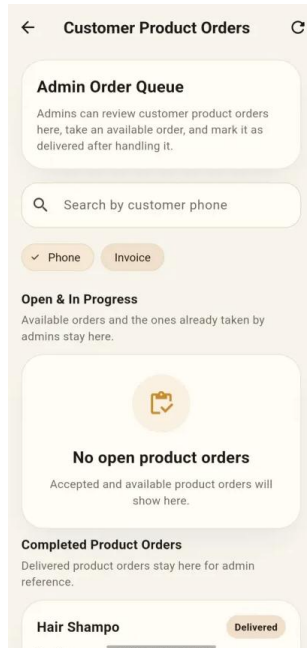


Figure 6.5-14. Customer order Screen

The Customer Orders window enables managers to monitor and manage customer orders for products. The Customer Orders window is illustrated in Figure 6.5-14. This window provides a summary of both pending and completed orders along with other relevant information.

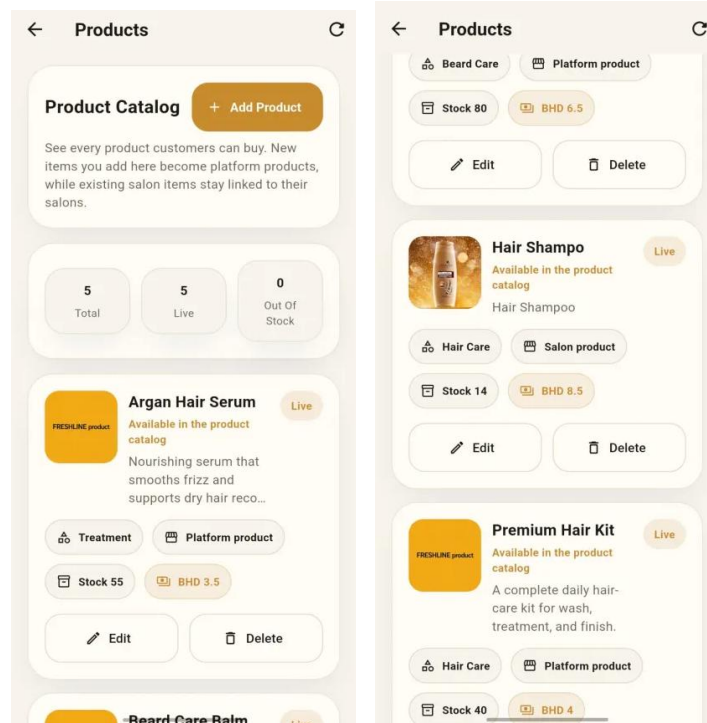


Figure 6.5-15. Products live screen

The products screen gives the administrator control over the platform's products, as depicted in Figure 6.5-15 below. This screen provides information on the product's name, category, cost, and inventory status, together with buttons for adding, modifying, or removing a product from the list.

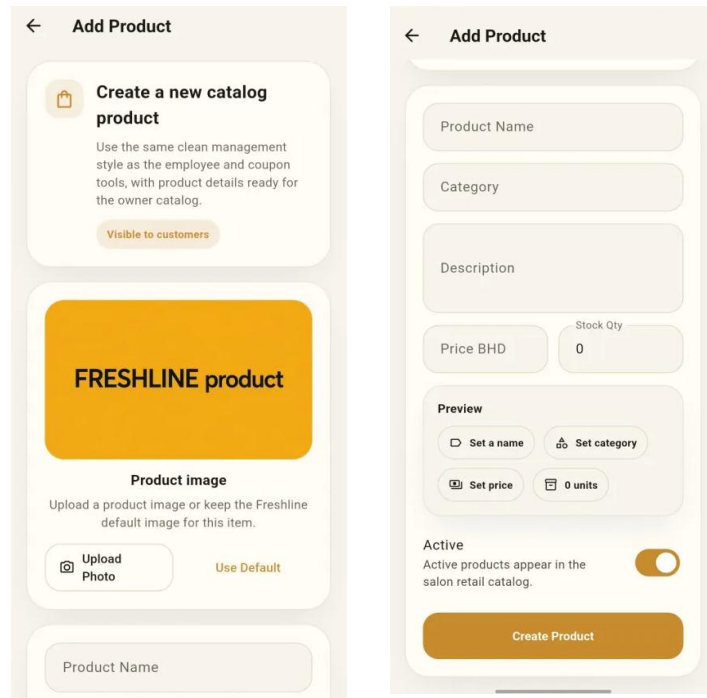


Figure 6.5-16. Add Products

The add product page helps the administrator to add new products to the platform's product catalog. This page has all the necessary fields that will enable you to enter information about your product, including the product name, category, description, price, and stock. You can even add the image of the product to the catalog using this page.

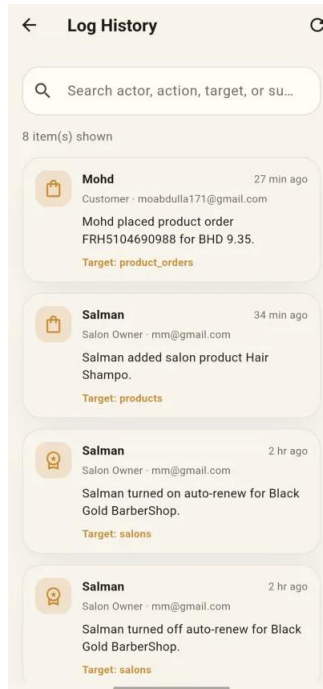


Figure 6.5-17. Log History

The log history window provides a way for the administrator to keep track of the actions that occur within the system, as depicted in Figure 6.5-17. This window displays the log histories of activities like placing an order, updating products, or making any modifications to the system.

6.6 Customer Side

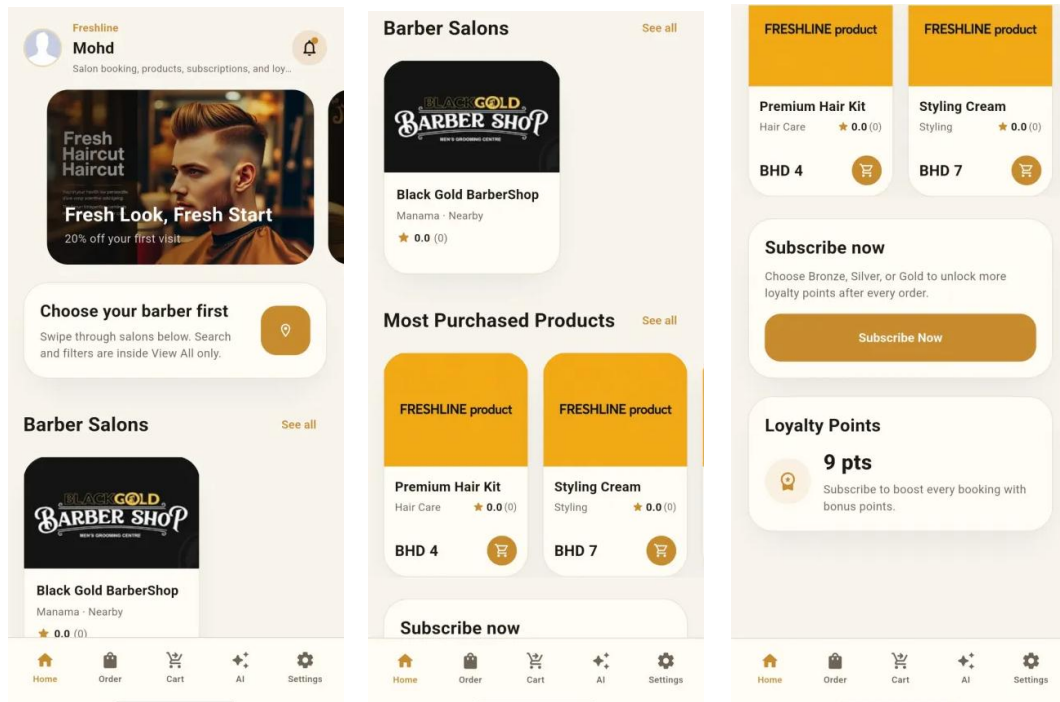


Figure 6.6-1. home screen

The home screen of the customer's section gives the users a brief idea about the salons available, the products, and the discounts offered, as can be seen in Figure 6.6-1. The home screen highlights popular salons, products, subscription plans, and loyalty points.

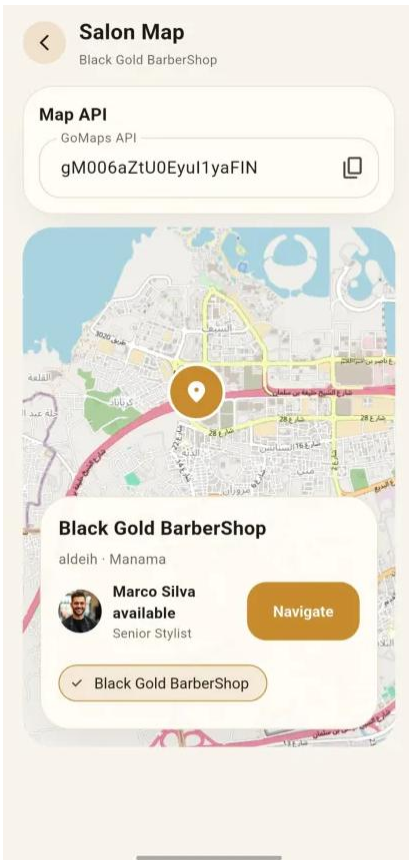


Figure 6.6-2. navigations all salon

The map of salons feature enables users to locate their required salon by viewing the map through which they can navigate to it. This is shown in Figure 6.6-2 where users can get details about the chosen salon and navigate to it.

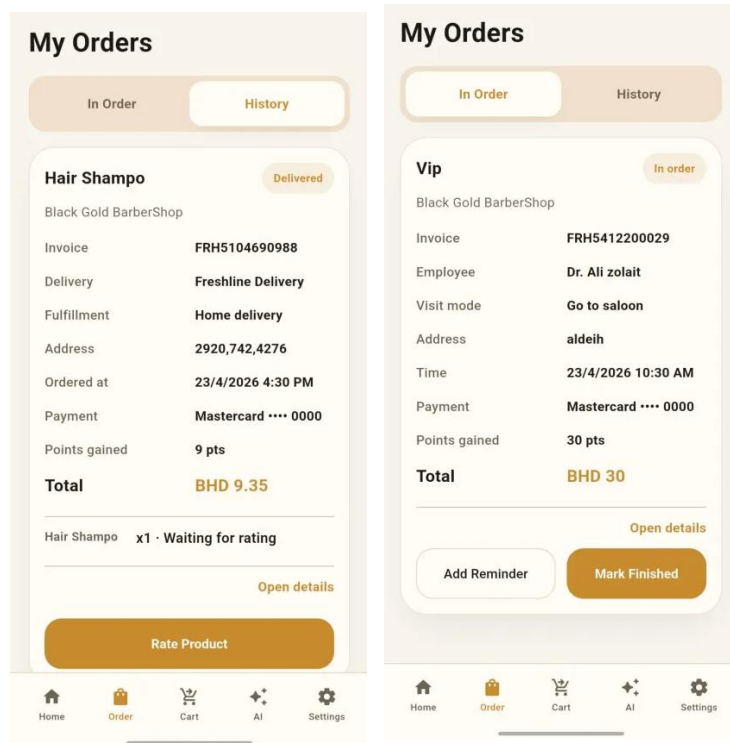


Figure 6.6-3. order screen

The Orders page enables the clients to view all their orders and bookings, as illustrated in Figure 6.6-3 below. The page is equipped with all past and existing orders that show their status, payment, and service details, among others.

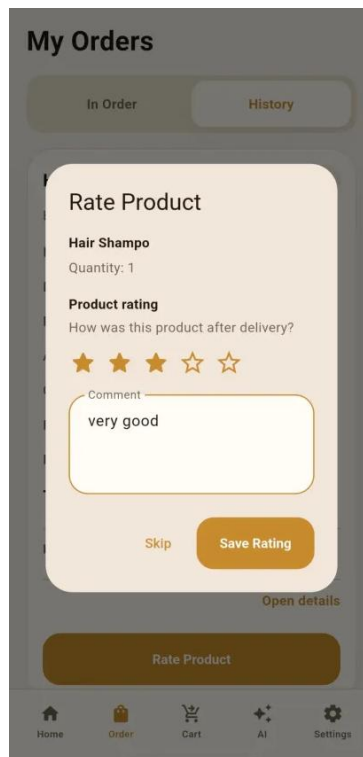


Figure 6.6-4. Rate order

Customer ratings of products can be seen using the rate product screen, as illustrated in Figure 6.6-4 below. It consists of the rating process together with a comment area, where users can give their feedback about the product upon delivery.

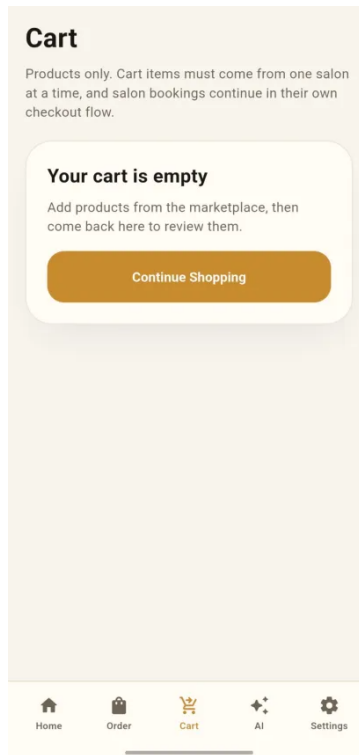


Figure 6.6-5. Cart screen

Cart screen helps the customer view chosen items before paying for them, as seen in Figure 6.6-5. It shows cart information and gives a choice of returning to the store when there is nothing in the cart.

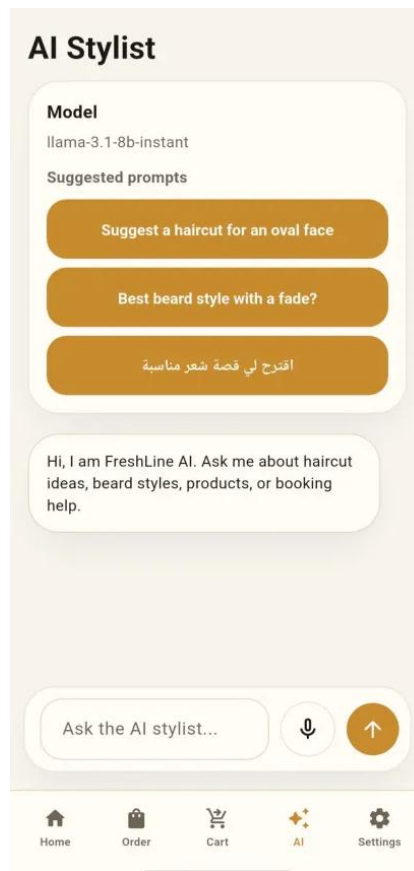


Figure 6.6-6. Ai for customer screen

The screen with the AI stylist lets clients communicate with an intelligent assistant for styling tips, as can be seen in Figure 6.6-6. The screen features prompts and a chat box that clients can use to ask questions about hairstyles, beards, and styling products.

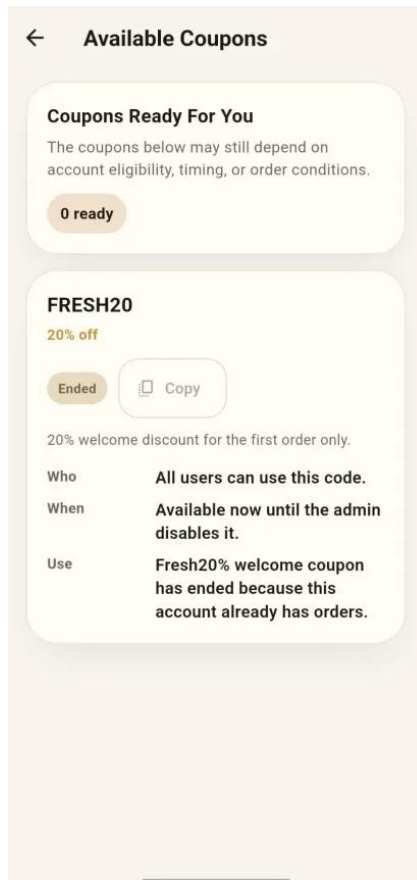


Figure 6.6-7. Coupon available

Coupon screen displays all the coupons available to the users, which are used with the coupon codes as shown in figure 6.6-7. Information about the coupon is displayed, such as the amount of discount, status, and its uses.

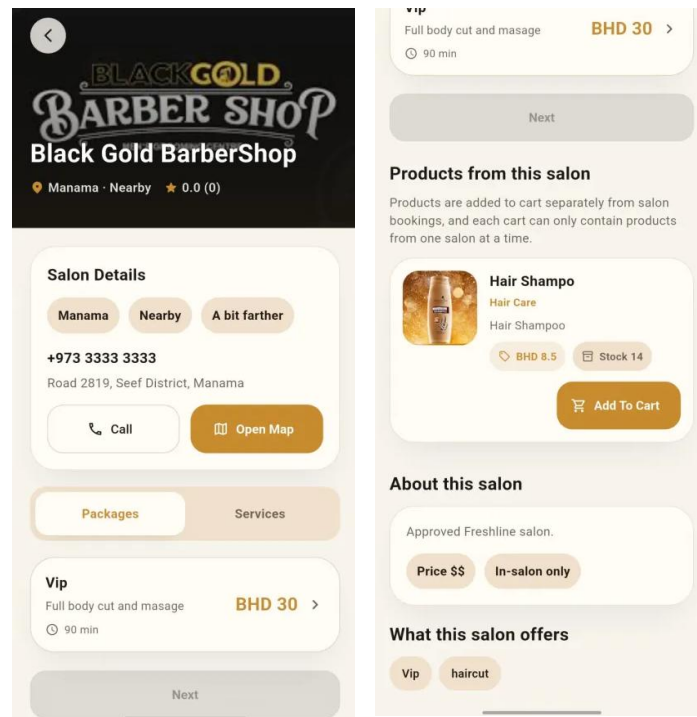


Figure 6.6-8. Salon Details

Customers have the ability to see the details of any selected salon on the Salon details page, as illustrated in Figure 6.6-8. The page provides details about the salon including location, contact methods, services, packages, and products offered at the salon.

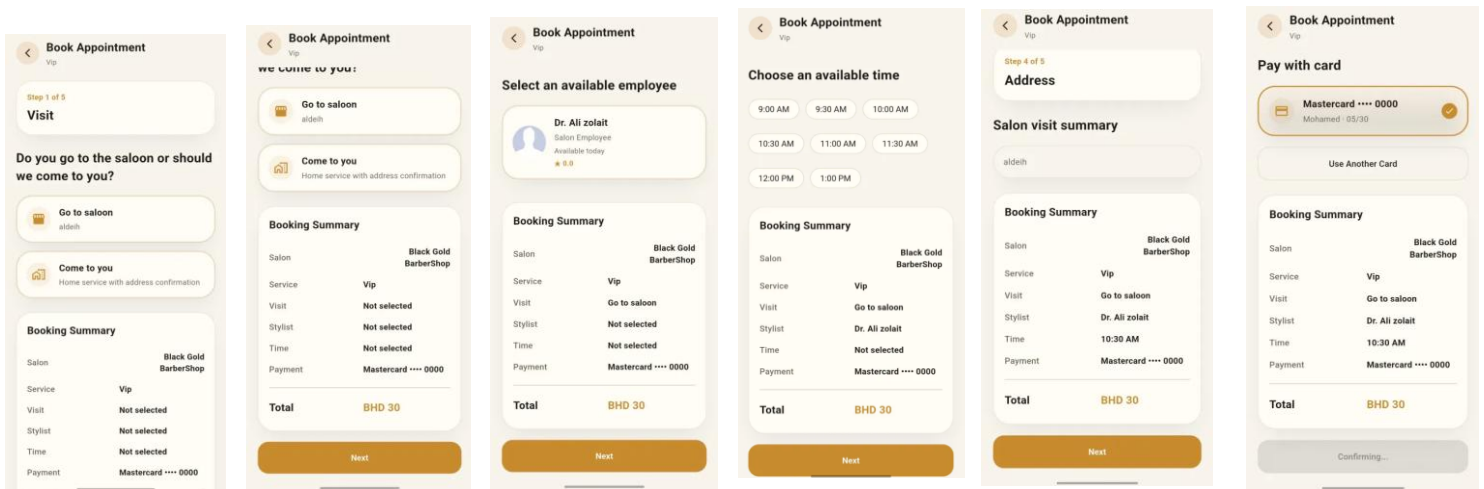


Figure 6.6-9. Step to book screen

Booking procedures serve as guidelines that help the customers go through the processes of making the appointment, which is illustrated in Figure 6.6-9. These processes entail the selection of visit types, selecting the appropriate staff, picking the right time, inputting address information, and making payments.

Chapter 7

System Implementation and Testing

7.1 Introduction

This chapter describes the implementation and testing of the Freshline system, an innovative mobile application designed to assist with booking salons and grooming services. This chapter describes the implementation of the application, its design, and various integration processes of the system elements to form a full-featured system based on selected technologies. Besides, this chapter provides evaluation of the proposed system in terms of its testing.

Testing of the application involves usability and user experience evaluation, and it is accomplished through the use of real users that have used the system and evaluated it in accordance with the questions listed in a special questionnaire. Evaluation outcomes provide information about the effectiveness of the system in question.

7.2 System Implementation

7.2.1 Frontend Development

The Freshline app's front end was implemented using flutter which is a cross-platform tool that allows developing mobile applications for the Android and iOS operating systems from a single code base. This tool was selected for its flexibility, faster development time, and ability to create highly interactive interfaces.

The user interface of the application was made user-friendly and easy to use with the possibility of switching from one feature to the other without any learning curve, be it searching salons, booking appointments, buying products or managing accounts.

7.2.2 Backend Development

The backend of this system was designed using Supabase which is a cloud-based platform that delivers database management, authentication, real-time capabilities, and other functions. Supabase was chosen for its robust and secure way of managing application data.

This platform is used to verify users, maintains information on salons and bookings, orders products, and chat in real-time.

7.2.3 System Integration

Integration of the frontend and backend was done using APIs and real-time communication facilitated by Supabase. With this, the system will be able to transfer data easily and effectively, ensuring that everything is connected perfectly.

All user interactions including booking any of our services and making an order will be instant, since everything will be stored and processed on the backend. Moreover, notifications and other updates are transferred in real-time too.

7.3 Testing Approach

1. What is your age?

[More details](#)

● Under 18	3
● 18–24	8
● 25–34	4
● 35–44	3

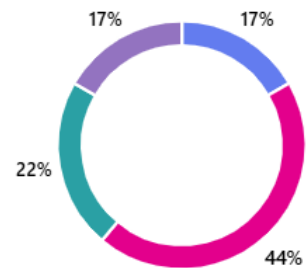


Figure 7.3-1: Age Distribution of Participants

As shown in Figure 7.3-1, the majority of participants (8 users, 44%) are aged between 18–24, followed by 25–34 (4 users, 22%), while both under 18 and 35–44 groups each represent 3 users (17%).

2. What Is Your Gender?

[More details](#)

● Male	13
● Female	5

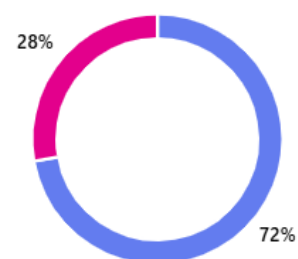


Figure 7.3-2 Gender Distribution of Participants

As shown in Figure 7.3-2, the majority of participants are male (13 users, 72%), while female participants represent 5 users (28%).

3. How often do you visit a salon or barber?

[More details](#)



Figure 7.3-3: Frequency of Salon Visits

As shown in Figure 7.3-3, most participants visit a salon weekly (8 users, 44%), followed by every 2–3 weeks (5 users, 28%), while monthly visits account for 3 users (17%) and rarely for 2 users (11%).

4. Did you use the Freshline application successfully?

[More details](#)



Figure 7.3-4: Successful Use of the Application

As shown in Figure 7.3-4, all participants (18 users, 100%) reported that they were able to use the Freshline application successfully, indicating a high level of usability and functionality.



Figure 7.3-5: Issues Faced by Users

As shown in Figure 7.3-5, no participants reported any issues while using the Freshline application indicating that the system operated smoothly during testing.

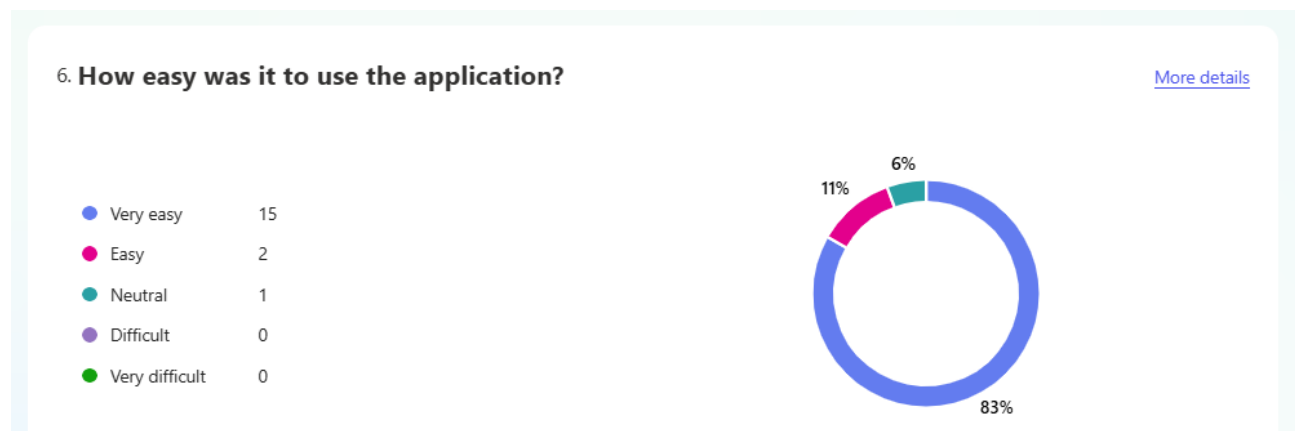


Figure 7.3-6: Ease of Use of the Application

As shown in Figure 7.3-6, the majority of participants rated the application as very easy to use (15 users, 83%), followed by easy (2 users, 11%), and neutral (1 user, 6%), with no users reporting difficulty.

7. Which parts were difficult to use?

[More details](#)

Booking process	9
Navigation	3
Product shopping	1
Payment process	4
None	7

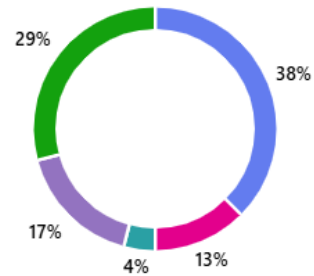


Figure 7.3-7: Difficult Parts of the Application

As shown in Figure 7.3-7, the booking process was reported as the most difficult part by 9 users (38%), followed by the payment process with 4 users (17%) and navigation with 3 users (13%), while product shopping was selected by 1 user (4%), and 7 users (29%) reported no difficulties.

8. Which features did you use?

[More details](#)

Booking services	15
Browsing salons	12
Buying products	10
AI assistant	11
Notifications	8

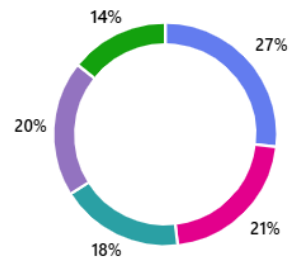


Figure 7.3-8: Features Used by Participants

As shown in Figure 7.3-8, booking services were the most used feature (15 users, 27%), followed by browsing salons (12 users, 21%), AI assistant (11 users, 20%), buying products (10 users, 18%), and notifications (8 users, 14%).

9. How satisfied are you with the app features?

[More details](#)

Very satisfied	13
Satisfied	3
Neutral	2
Unsatisfied	0

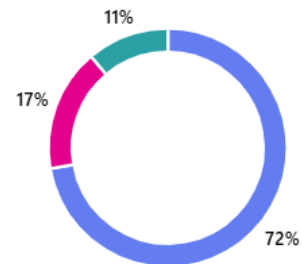


Figure 7.3-9: User Satisfaction with Application Features

As shown in Figure 7.3-9, most participants were very satisfied with the application features (13 users, 72%), followed by satisfied (3 users, 17%) and neutral (2 users, 11%), with no users reporting dissatisfaction.

10. How was the app performance (speed & response)?

[More details](#)

Excellent	14
Good	1
Average	3
Poor	0

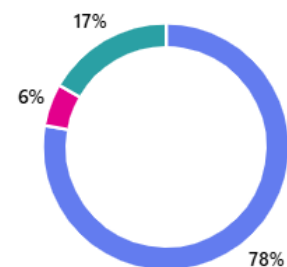


Figure 7.3-10 Application Performance (Speed and Response)

As shown in Figure 7.3-10, most participants rated the application performance as excellent (14 users, 78%), followed by average (3 users, 17%) and good (1 user, 6%), with no users reporting poor performance.

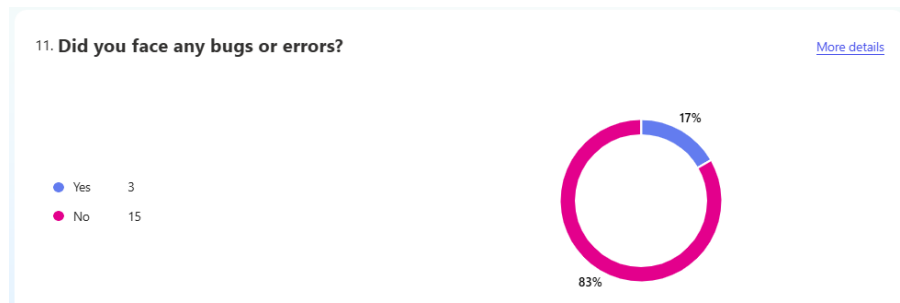


Figure 7.3-11 Bugs or Errors Encountered

The majority of the participants, 15 users (83%), reported not encountering any bugs or errors while using the application (as shown in Fig 7.3-11). Only 3 users experienced them.



Figure 7.3-12: Reported Issues or Bugs

As indicated in figure 7.3-12, only 3 participants stated any issues or bugs while the rest did not specify any concrete issues, meaning that the system was stable for most users.

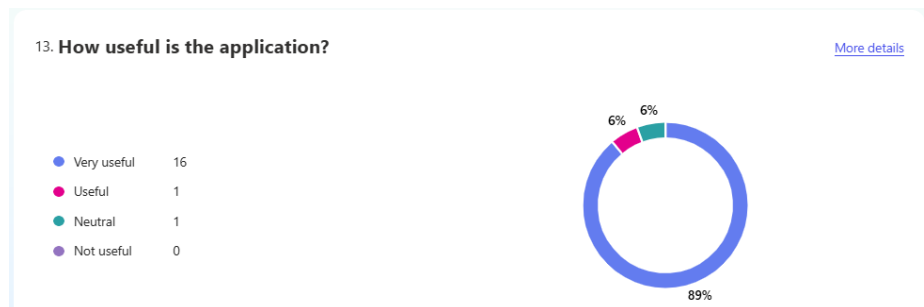


Figure 7.3-13: Usefulness of the Application

Figure 7.3-13 indicates that most users rated the application very useful, while one user (6%) rated it useful, and another one user (6%) remained neutral, with no user rating it not useful.

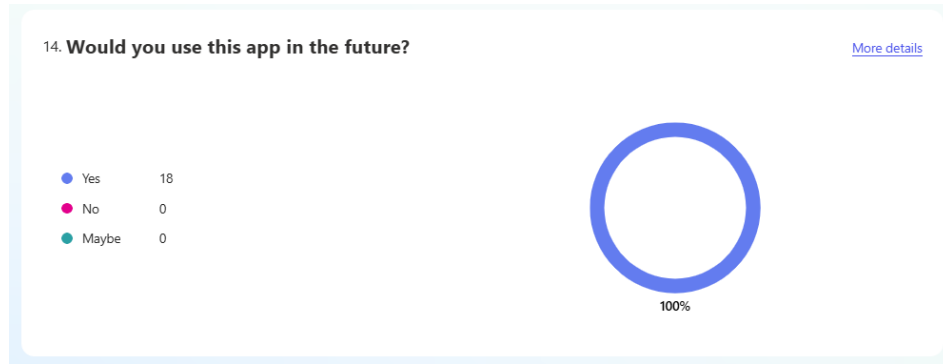


Figure 7.3-14 Future Usage Intention

Figure 7.3-14 showed that all the participants (18 users, 100%) stated that they could see themselves using the application in the future.

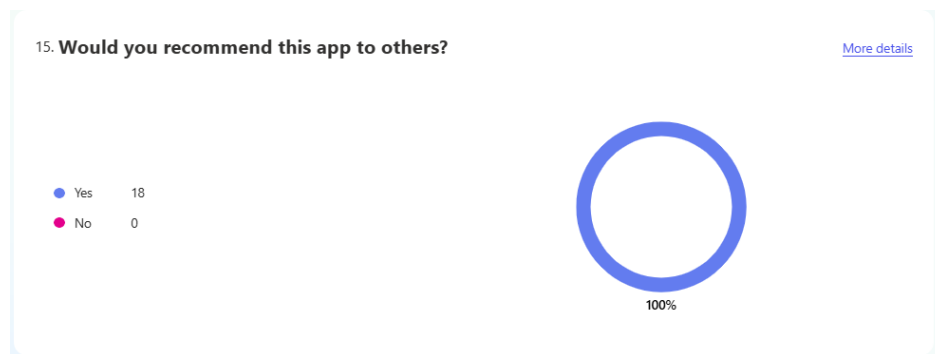


Figure 7.3-15: Recommendation of the Application

Figure 7.3-15 shows that the results from all participants, who are 18 users (100%), and said that they would recommend this application to people.

16. What improvements would you suggest?

18 Responses

ID ↑	Name	Responses	Language
1	anonymous	Thank you	English (United States)
2	anonymous	No everything being clear	العربية
3	anonymous	Keep it up	English (United States)
4	anonymous	Nothing	English (United States)
5	anonymous	No thing	English (United States)
6	anonymous	Very good	English (United States)
7	anonymous	Nothing	English (United States)
8	anonymous	More Services	English (United States)
9	anonymous	Good app	English (United States)
10	anonymous	Nothing	English (United States)
11	anonymous	Very good app	English (United States)
12	anonymous	Not a thing	English (United States)

Figure 7.3-16 Suggested Improvements by Users

Figure 7.3-16 indicates that the majority of participants had no substantial suggestions for improvement. “Nothing” and “Very good” were the most common responses. A few users suggested expanding the range of services, a minor area of improvement, though overall satisfaction is high.

7.3.1 Questionnaire Design

This questionnaire aimed to test the usability and user experience of Freshline application post-interaction with the software. A total of 16 questions made up this survey, with all of them testing crucial elements such as ease of use, navigation through the software, system speed, usage of functions and features, among other factors. The questions were straightforward and easy to answer so as to guarantee relevant feedback on how the system performed from a user perspective.

7.4 Strengths and Weaknesses of the System

Freshline shows good performance and usability since the program provides its users with ease of use, quick responses from the application, and integration of three different systems such as booking of salons, purchases of beauty products, and help provided by AI technology in one program. It is possible to assume that all the users could handle the system and perform their tasks smoothly because of the efficiency of its development and functioning. Nevertheless, there are several drawbacks to the system in terms of its availability and need for improvement in some respects in order to make it more user-friendly.

7.5 Conclusion

The implementation and testing of the Freshline app, a mobile application that allows customers to make appointments at the salons and groom pets, reveals that the app successfully accomplishes its aims efficiently. Developers employed the use of the latest technology like Flutter along with Supabase to build an application with countless options. The user testing shows that the app performed well in terms of usability and user experience.

Chapter 8

Conclusion and Future Work

8.1 Introduction

The project involved development of a live mobile app, which gave us individual experience in developing a complete digital solution and managing its deployment. Through Freshline we overcame the existing problems in the salon and grooming business with a consolidated solution by using different technical skills. The project calls for designing mobile app's frontend using Flutter and backend with Supabase as per client requirements.

While working on the project, several difficulties were faced, such as different user types, system integration, effective user experience, and so on. Through tackling these difficulties, we managed to improve our problem-solving and teamwork skills, plus our knowledge of modern software development approaches.

8.2 Conclusion

Freshline has successfully created a smart salon booking and grooming system which is more effective than the traditional salon services. The application is user-friendly for both the consumer and the business owner as it combines multiple functions into one platform such as appointment scheduling, products purchase, AI assistant and business management. Obstacles the team faced during the design included the need to accommodate multiple user roles, as well as the backend and frontend interfaces able to integrate effectively.

To efficiently execute the system, the technology of Flutter and Supabase is used. The actual users were involved in contacting testing procedure to verify the usability and efficiency of the app. To conclude, this project helped to improvise various skills and also helped to work together. Besides, it depicted the necessity of taking consumer feedback and further improving the product.

8.3 Future Work

Freshline's tomorrow will be all about user-centric smart products. Efforts will then be made to enhance the end-user experience. For instance, one such enhancement will be the addition of a virtual try-on feature that will allow the user to choose a particular hairstyle or grooming service and see how it looks like before they book the session.

Another smart feature that will be introduced in Freshline will be the implementation of a high-end AI agent. Using this agent, users will just have to give simple instructions by requesting the agent for a service and the agent will automatically pick the appropriate salon and make a booking accordingly.

In addition, in future works, a web application will be developed with the mobile application, giving access to the user and salon owner for use in multiple devices. Another important aspect will be scaling operations of Freshline to the GCC countries.

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